Rules versus Discretion: A Collateral Valuation Reform in Government-Guaranteed Small Business Lending*

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Abstract

Lender discretion over collateral valuation encodes a private read on what business assets would recover in default. A January 2014 SBA reform replaced that discretionary appraisal with a preset, uniform haircut schedule, erasing the read without changing what borrowers can pledge. The reform's first-order effect is not a repricing but an adaptation in how lenders take collateral: where home equity offers no recovery buffer, lenders rebuild the lost read with costlier, more asset-specific liens on the movable collateral they keep, raising collateral specificity by about 0.18 of a standard deviation in the three states with classified collateral filings; the accompanying shift away from real estate, the direction the adaptation predicts, is more weakly identified. Prices move only as a consequence: spreads on new borrower-lender matches fall about 10 basis points, the muted response a 75 percent guarantee predicts for a valuation-only reform, graded by the state homestead exemption. The lender's relationship read on the borrower is untouched, so holding the borrower and lender fixed spreads show no detectable shift, consistent with the gradient being a cross-sectional new-borrower phenomenon.

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1 Introduction

A recurring question in financial regulation is whether to constrain lender discretion. Private judgment lets lenders draw on information about borrowers and their collateral that outside parties cannot observe (Berger and Udell, 1995; Petersen and Rajan, 1994), but it also produces conservative bias, rationing, and inconsistency (Stiglitz and Weiss, 1981; Manove et al., 2001). When a regulator replaces lender judgment with a rule, the private information embedded in that judgment is destroyed along with its costs (Liberti and Petersen, 2019). Three questions follow. What private information does the rule erase? Can lenders rebuild what the rule took away? And how does the price of credit respond? I answer these questions for a reform that standardized how collateral is valued.

The collateral literature has largely settled *what* can be pledged and under *what legal rules*: which assets are pledgeable (Ongena et al., 2025), how redeployable they are (Benmelech and Bergman, 2009), and how legal reforms expand the set of assets a lender can claim (Calomiris et al., 2017; Campello and Larrain, 2016; Aretz et al., 2020; Vig, 2013). A different lever is the value a lender assigns to a fixed set of pledged assets. The closest antecedent exploits an exogenous fall in that value: when a Swedish reform stripped floating liens of priority, lenders raised loan rates and monitored less (Cerqueiro et al., 2016). I study a reform that works through the same value channel in mirror image, standardizing the lender's discretionary *valuation* of an unchanged, already-pledgeable asset set. Existing work measures how much collateral is worth under lenders' discretionary appraisal (Luck and Santos, 2024); I study what happens when a regulator removes that discretion. Holding fixed which assets are pledgeable and which legal rules govern them, a regulator can standardize how those assets are *valued*, replacing the lender's private appraisal with a uniform formula. Standardizing valuation strips a particular kind of private information, the lender's own read on how much a given pool of business assets would fetch in default, while leaving every other margin of lender knowledge intact. This paper asks what happens to pricing, lender conduct, and credit allocation when the valuation method on already-pledgeable assets is standardized.

The setting is a January 2014 reform of the Small Business Administration's 7(a) program. Before the reform, lenders set the recognized value of business collateral using their own discretionary appraisal, an "orderly liquidation value" determined case by case. The reform replaced that appraisal with a preset, uniform schedule of haircuts on net book value, identical across lenders and loans. The change was procedural and applied to every 7(a) lender at once, which makes it a clean shock to the valuation method without changing which assets borrowers could pledge or the legal rules that perfect those claims.

A uniform reform offers no cross-sectional comparison on its own, so I use the state homestead exemption as a moderator of exposure rather than as the treatment. The logic runs through how a lender recovers in default. Recovery is structured in two stages. A lender first looks to the business collateral it has perfected; to the extent that falls short, it can pursue a deficiency claim against the borrower's other personal assets. The homestead exemption governs the second stage: where the exemption is large, the borrower's home and related personal assets are shielded from the deficiency claim, so recovery leans almost entirely on the business collateral seized in the first stage. The discretionary appraisal the reform standardized was the lender's private valuation of exactly those first-stage assets. Its pricing weight is therefore largest where the home is most shielded, which is to say where recovery concentrates on business collateral. Because the reform applies uniformly and the exemption grades exposure to it through recovery structure, there is no selection into treatment: lenders did not choose their state's exemption in response to the reform, and the exemption predates it by decades.

This recovery structure organizes the paper around two private-information margins with two fates. One margin is the lender's read on the recovery value of an entire asset class. The reform takes this directly, and it is recoverable: a lender can rebuild an asset-class read by recording the underlying assets more precisely in its collateral filings. The other margin is the lender's relationship knowledge of a specific borrower. The reform never touches this, so it stays, and its persistence shows up in the absence of any detectable repricing within an ongoing borrower-lender relationship. The reform's first-order effect is the lender's rebuilding of the asset-class read, an adaptation in how collateral is taken; the change in spreads is a downstream consequence of that adaptation, surfacing in the cross-section of newly formed matches rather than in within-borrower repricing.

I develop three predictions. *First*, lenders adapt the collateral they take: where they cannot fall back on home equity they rebuild the asset-class read the reform took away by filing more detailed, asset-specific descriptions, while where home equity provides a recovery buffer they take the cheapest route to perfection and file generic blanket liens. *Second*, this adaptation shows up in prices: the reform lowered spreads most where business collateral is the lender's main recovery channel, so the more a state's homestead exemption shields the borrower's home and other personal assets, the larger the spread reduction. *Third*, for a borrower who stays with the same lender across the reform, the spread does not change, because an incumbent's price is disciplined by what a competitor would offer and the competitor prices off pooled borrower-type information the reform leaves untouched, so the gradient is a cross-sectional new-borrower phenomenon rather than within-borrower repricing.

I observe this adaptation directly by matching SBA loans to Uniform Commercial Code filings in California, Florida, and Colorado, the three states for which a UCC collateral classification is available; Florida has an unlimited homestead exemption while California and Colorado have limited exemptions, so the three-state sample spans the exposure variation within the subset of states where UCC data exist. Where lenders cannot fall back on home equity, they file more detailed, asset-specific descriptions: in unlimited-exemption states filing specificity rises by about 0.18 of a standard deviation, and a composite filing-intensity index moves in the same direction. Where home equity provides a recovery buffer, they take the cheapest route to perfection instead, and filing rose by 13.8 percentage points more in the limited-exemption states, overwhelmingly in generic blanket-lien language. I read the added detail as the lender rebuilding the asset-class read the reform erased, a revealed-preference interpretation the later filing-mix evidence is consistent with rather than a directly observed intent; the loan-level match of SBA 7(a) loans to their UCC filings is, to my knowledge, the first of its kind. As a descriptive companion, the fraction of loans requiring any collateral fell by 4.9 percentage points more in unlimited-exemption states, so the reform reshaped what collateral lenders take, not only how they price it.

Prices move as a consequence of this adaptation. I grade cross-state exposure by the effective homestead exemption in a borrower's state, the larger of the state and federal exemption caps. Across all 49 jurisdictions (103,476 loans in the preferred specification), moving from the lowest effective exemption to an unlimited-exemption state lowers spreads by about 10.2 basis points; for the median treated-state loan ($1.17 million), a reduction of this order is roughly $1,250 in annual interest savings, and about $19 million per year across treated-state loans. The per-loan magnitude is modest against an average spread near five percentage points, the muted response expected when the 75 percent federal guarantee absorbs most of any revaluation. The continuous gradient is sensitive to Texas, the one state where a constitutional bar makes the lender's dependence on business collateral exact; excluding all seven unlimited-exemption states, the gradient remains significant within the interior of the distribution, so the design-robust effect lives in the interior of the exemption distribution rather than in the Texas corner. It is stable across loan-size bins within the $350K+ tier, which rules out the July 2014 credit-scoring screen below that threshold. The new haircuts raised recognized collateral value on average even as they removed the lender's private valuation advantage, because the pre-reform orderly-liquidation-value baseline was conservative by design.

The within-borrower evidence sorts the two margins. Holding the borrower fixed, the spread for repeat borrowers who stay with the same lender shows no detectable change, and the estimate is too imprecise to rule out repricing of around twenty basis points in either direction, so I read it not as an affirmative within-borrower zero but as consistent with the relationship-information account: an incumbent's price is disciplined by what a competitor would offer, and the competitor prices off pooled borrower-type information the reform leaves untouched. The pricing gradient is identified across borrowers, so by design it is a cross-sectional new-borrower phenomenon, and the underpowered within-borrower test shows no offsetting movement inside ongoing relationships. The reform's pricing effect is therefore consistent with operating through the formation of new borrower-lender matches, on who newly lends to whom, rather than through the repricing of ongoing relationships.

Two conditions bound external validity. The mechanism evidence on collateral specificity is identified on the three-state California, Florida, and Colorado UCC sample for which classified collateral filings are available, and the pricing consequence is estimated on Standard Guaranty loans above $350K, where the guarantee rate and collateral adequacy rule are uniform. The adaptation evidence is direct, mechanism-level evidence on three states with a single treated cluster, Florida; I read it as co-equal with, not more robustly identified than, the 49-jurisdiction pricing design. These responses reflect changes in lender conduct, not in borrower selection or screening. Falsification tests on charge-off rates, state-quarter loan counts and lending volume, lender entry, the repeat-borrower share, and an Express-program placebo all show no effect, consistent with the reform changing valuation methodology rather than the composition of borrowers or the quality of screening. Standard errors are clustered at the lender level throughout.

The paper makes three contributions. *First*, it isolates the valuation-method margin. Holding pledgeability, the legal rules, and the pledged asset set all fixed, I standardize only how already-pledgeable assets are *valued*. This is the mirror image of a value-lowering collateral shock, which raises loan rates and slackens monitoring (Cerqueiro et al., 2016): here the standardized schedule raises recognized value, so spreads fall rather than rise, and lenders respond by producing information rather than withdrawing it. Where prior reforms changed which assets a lender can reach (Calomiris et al., 2017; Ongena et al., 2025), I change only how the reachable assets are valued, showing that the assessment method reshapes lender conduct first and, through it, pricing. It connects to the view of secured debt as an enforcement technology (Rampini and Viswanathan, 2025): where that work studies enforcement under given collateral values, I make the valuation itself a regulatory parameter and trace how standardizing it reshapes the lender's selection and enforcement. It also connects to lender-based theories of private information about borrowers (Inderst and Mueller, 2007), whose channel linking a narrower information advantage to greater collateral use is the very margin my restriction to loans above $350K removes, so the specificity response I estimate reflects the valuation-method change rather than a screening-technology one. *Second*, the collateral-specificity response runs opposite to what regulatory standardization usually elicits, and it follows from the information logic of how lenders take collateral. A floating (blanket) lien[1] attaches to a replaceable pool of business property, so firm-level financials already speak to its recovery value; an asset-specific pledge fixes on identified assets the borrower cannot dispose of without consent, and firm-level financials do not convey the value of those individual pledged assets, so the lender must produce collateral-specific information itself. Where the home cannot backstop recovery, a lender whose claim has shifted toward specific pledged assets must rebuild the asset-class read the rule erased, and the detailed, asset-specific filings I observe are consistent with that information-production response. This is the prediction the mechanism makes, not merely a pattern read after the fact: when regulators standardized risk weights, banks manipulated the discretionary inputs to lower their capital charges (Behn et al., 2022; Plosser and Santos, 2018), and intermediaries responded heterogeneously to mandated standardization in mortgage renegotiation (Agarwal et al., 2017); here lenders cannot manipulate a preset haircut, so where recovery depends on business collateral they instead invest in observable, asset-specific filings to reconstruct the private read the rule erased. The detailed filings are observed behavior consistent with information rebuilding rather than directly observed intent. *Third*, the exemption gradient is, by design, a cross-sectional new-borrower phenomenon: the gradient is identified across borrowers, so the reform's pricing effect operates through the formation of new matches rather than through within-borrower repricing. Because an incumbent's relationship information about a specific borrower is reform-invariant while a competing entrant prices off pooled information the reform also leaves untouched, the incumbent's advantage neither widens nor narrows in price for an ongoing relationship; the pricing response surfaces among borrowers who enter the market anew after the reform. The borrower-fixed-effects spread estimate is small and shows no detectable within-borrower movement, consistent with this reading, though it is too imprecise to affirm within-borrower invariance on its own.

The paper also speaks to two adjacent literatures. On lending relationships (Petersen and Rajan, 2002; Liberti and Petersen, 2019; Blickle et al., 2025), the null charge-off result rules out a screening-quality explanation for the spread change and locates the effect in pricing rather than risk. On public collateral registries, Mann (2018), Gopal and Schnabl (2022), and Gopal (2021) use USPTO assignments and UCC filings to read lender behavior; I bring that approach to SBA lending. Finally, on homestead exemptions and small

[1] The empirical measures that quantify a filing's position between the floating-lien and specific-pledge poles, the blanket-scope indicator, the scope score, and the specificity score, are defined in the data section.

business credit (Gropp et al., 1997; Berkowitz and White, 2004; Cerqueiro et al., 2017; Ersahin et al., 2021; Cole et al., 2025), prior work treats the exemption as the treatment; I use it to grade exposure to a uniform reform, which removes selection into treatment. On government-guaranteed lending (Brown and Earle, 2017; Bachas et al., 2021; Collier et al., 2025; Pan et al., 2025), I show how the rules for valuing collateral shape pricing, allocation, and conduct, complementing Hackney (2023) on the 7(a) program's institutional structure.

The remainder of the paper proceeds as follows. Section 2 describes the SBA 7(a) program and the January 2014 collateral valuation reform. Section 3 develops a selection-and-enforcement framework for how lenders rebuild the asset-class read once valuation discretion is withdrawn. Section 4 describes the data and sample construction, including the SBA-to-UCC match. Section 5 lays out the empirical strategy. Sections 6 to 7 report the lender adaptation evidence and the pricing consequence. Section 8 discusses the new-match interpretation of the pricing gradient, alternative interpretations, and policy implications. Section 9 concludes.

2 Institutional Background

2.1 The SBA 7(a) Loan Program

The SBA 7(a) program is the largest federal small business lending program, approving \$15–\$25 billion annually across 40,000–60,000 loans during the sample period (2011–2019).[2] The SBA does not lend directly; it guarantees 75% of each loan in the sample universe, with the lender bearing the remaining 25% of any net loss after collateral liquidation. The lender makes the credit decision and sets the interest rate; the SBA reviews the file and authorizes the guarantee. The lender's collateral valuation methodology therefore directly shapes expected recovery on its retained share and the spread it charges.

In addition to business collateral, the SBA requires a *personal guarantee* from all owners with 20% or greater ownership. The personal guarantee is an unsecured legal obligation (a mandatory program requirement, not a negotiated term) that exposes the guarantor's personal assets to creditor claims upon default (Berkowitz and White, 2004). The practical reach of this guarantee depends on the state homestead exemption, which the next subsection introduces.

Collateral is required "to the maximum extent possible." For loans above \$350K, the lender must first take security interests in available business assets; only if a shortfall re-

[2] Author's calculations from the SBA 7(a) loan-level disclosure files. Program requirements are set forth in 13 CFR Part 120 and the SBA's Standard Operating Procedure 50 10 (Chapter 4).

mains is the lender required to take available equity in the personal real estate of the principals. This requirement is unchanged by the reform and applies throughout the sample period. Whether the loan is deemed "fully secured" depends on the valuation methodology applied to these assets, the subject of the January 2014 reform.

2.2 Collateral Valuation Before 2014

Before January 2014, the SBA's collateral provisions specified no asset-type-specific valuation rules. A loan counted as "fully secured" when the lender held security interests in assets whose combined *liquidation value* reached the loan amount, where liquidation value was the amount the lender expected to realize by taking possession after default and selling the asset after a reasonable search for a buyer, net of disposition costs and prior liens.[3] This delegated the entire valuation to the lender. The collateral rules provided no asset-type-specific haircuts, no reference schedule, and no formula linking asset characteristics to collateral value. "Liquidation value"[4] was whatever the lender judged it to be: an estimate of hypothetical proceeds under hypothetical default conditions, net of hypothetical costs. Two lenders evaluating identical equipment could arrive at different liquidation values, and the rules imposed no constraint on the gap.

The only administrative check on this discretion was a nominal appraisal requirement for high valuations, which in practice certified rather than restrained the lender's judgment. A relationship lender familiar with the borrower's maintenance practices, the local secondary market for the equipment, or the borrower's operating track record could select an appraiser whose methodology was compatible with the lender's assessment, rendering the appraisal requirement a procedural rather than substantive constraint. The assessed value of business collateral was therefore a lender-specific quantity, varying across lenders for the same borrower and across borrowers at the same lender depending on the depth of the relationship. This variation is the source of the lender's pre-reform discretionary appraisal, and it draws on two distinct kinds of private information. Asset-class knowledge (the local secondary market for specialized equipment, wear-and-tear patterns) feeds into the lender's read on what business collateral would fetch in default; relationship-specific knowledge of borrower type (operational history, management quality) feeds into the lender's read on the borrower's underlying risk. Both fed into the lender-specific orderly-liquidation-value practice, and the reform reaches only the first: standardization replaces the lender's discretionary appraisal of business assets with a preset haircut, while the re-

[3]SOP 50 10 5(E), p. 189.

[4]I use "orderly liquidation value" (OLV) for this pre-reform concept: an orderly-sale value, not a fire-sale.

lationship knowledge that prices borrower type is left untouched.

2.3 The January 2014 Collateral Standardization

Effective January 1, 2014, the SBA replaced discretionary liquidation value with a schedule of preset haircuts on net book value (NBV) for business assets. A loan now counted as "fully secured" when the combined adjusted net book value of available fixed assets, real estate, machinery, and equipment, reached the loan amount, where net book value is an asset's original price minus depreciation and amortization.[5]

The shift from a subjective "liquidation value" to "net book value as adjusted" is the core of the reform: a subjective valuation becomes an accounting quantity scaled by preset percentages. Table 2 reports the haircut schedule. The reform applied uniformly to all participating 7(a) lenders on January 1, 2014, with no phase-in, pilot program, or lender-class opt-out, so the entire lender population faced the same rule change on the same date. The SBA announced the new procedures on September 25, 2013, roughly one quarter before they took effect.[6]

The impact differs sharply by asset type. For **business assets** (machinery, equipment, furniture and fixtures), new equipment receives a mechanical 75% haircut on its purchase price and used equipment receives 50% of net book value, or 80% if the lender obtains an orderly-liquidation-value appraisal. Even this residual appraisal option operates within a preset range, unlike the unconstrained lender judgment of the pre-reform regime. A relationship lender who knew the borrower's equipment was well maintained can no longer translate that knowledge into an arbitrarily generous valuation. For **real estate**, by contrast, the reform introduced an 85% haircut on market value but left the underlying appraisal process untouched. Real estate appraisals remain inherently discretionary: the appraiser selects comparables, adjusts for condition and location, and exercises professional judgment. The 85% haircut standardizes the *discount applied to the appraisal* without standardizing the *appraisal itself*. Lender discretion over real estate valuation is therefore preserved.

Although the new haircuts eliminate lender-specific judgment, they raise recognized collateral value on average relative to the pre-reform regime. Pre-reform, the SBA could reduce or deny a guaranty repurchase claim, or impose a dollar-amount "repair," if the National Guaranty Purchase Center's post-default review found the lender's collateral identification, valuation, or documentation non-compliant with SOP 50 10 requirements

[5] SOP 50 10 5(F), p. 164.
[6] SBA Information Notice 5000-1290 (Sept. 25, 2013).

(13 CFR §120.524(a); SOP 50 57 Chapter 24). Contemporaneous SBA Office of Inspector General audits of early-defaulted 7(a) loans flagged "insufficient collateral," "inadequate appraisals," and "failure to prepare and use an adequate list of collateral" as named deficiency triggers,[7] and lenders bore the unguaranteed 25% residual loss share on any uncovered loss. Both pressures favored conservative discounts in expectation, so typical pre-reform haircuts were understood to sit well below the upper-bound percentages that the 2014 reform later codified. The shift to a preset-rule regime therefore raises the recognized baseline for most loans, even as it eliminates the relationship-specific upside that the pre-reform lender could extract through generous case-by-case valuations. Direct measurement of collateral values under lenders' discretionary appraisal corroborates this reading: such values are heterogeneous and often conservative, and an all-asset blanket lien is valued similarly to an unsecured claim (Luck and Santos, 2024), which is both why standardizing the appraisal raises the recognized baseline and why the generic blanket liens that home-equity-backstopped lenders default to carry little recovery value.

The estimation sample is restricted to Standard Guaranty loans above $350K precisely to avoid two contemporaneous policy changes that apply only to smaller loans.[8] The empirical strategy exploits cross-state variation in homestead exemptions to measure exposure to the business-asset valuation reform: lenders in unlimited-exemption states, who depend entirely on business collateral, are more affected than lenders in low-exemption states, who retain a home equity buffer above the effective exemption.

2.4 State Homestead Exemptions

Homestead exemptions are state-law provisions that protect a debtor's equity in a primary residence from creditor claims. They vary dramatically across states, from the lowest extreme (New Jersey and Pennsylvania, which provide no state-level homestead exemption at all) to unlimited protection (Arkansas, Florida, Iowa, Kansas, Oklahoma, South Dakota, Texas), with the remaining states spanning modest amounts of a few thousand dollars up to several hundred thousand dollars. This cross-state variation is the moderator exploited in the empirical design.

The binding protection is not the state statute alone. Under 11 USC §522(b)(2), twenty states and the District of Columbia let a Chapter 7 debtor elect the federal homestead

[7] E.g., SBA OIG Report 12-11R (2012) and Audit Report 13-07 (2013).

[8] The October 2013 fee waiver (loans $\leq$$150K) and the July 2014 credit-score screen (Standard Guaranty loans $\leq$$350K; SBA Notice 5000-1314) both apply only below the $350K threshold and so are excluded by the sample restriction. A contemporaneous Veterans Advantage fee waiver touches only SBA Express loans, relevant solely to the Express-program placebo.

exemption in lieu of a lower state amount, so in an opt-in state whose statute falls below the federal floor, about $23,000 for most of the sample, the shielding the lender faces is effectively the federal amount.[9] I use the *effective exemption* throughout, the greater of the state statute and the federal amount where the state permits the election, and the state statute otherwise.

The exemption matters for SBA lending not primarily through bankruptcy but through its effect on the lender's outside option in default. When an SBA borrower defaults and business collateral is insufficient to cover the net loss, the lender pursues the mandatory personal guarantee by seeking a deficiency judgment against the guarantor's personal assets. The effective exemption determines how far this claim reaches. In low-exemption states, the borrower's home equity above roughly $25,000 is exposed, providing the lender with a substantial additional recovery source that effectively extends the collateral base beyond the business. In unlimited-exemption states, the primary residence is shielded, and the deficiency judgment can reach only non-exempt personal assets (financial accounts, non-homestead real estate, vehicles above state limits), which for most small business owners are modest (Gropp et al., 1997; Berkowitz and White, 2004).

The economic consequence is a sharp difference in exposure to collateral valuation rules across the two regimes. In low-exemption states, the home equity buffer partially insulates expected recovery from the collateral reform. In unlimited-exemption states, recovery depends almost entirely on business assets whose valuation the reform standardizes, so the same policy shock generates a larger change in expected recovery and a larger pricing response. This asymmetry in exposure is the source of the identifying variation.

Among the unlimited-exemption states, Texas is unique: the Texas Constitution (Article XVI, Section 50) renders home equity liens for business purposes constitutionally unenforceable, so Texas borrowers cannot voluntarily pledge home equity even if they wish to.[10] The extreme-exemption classifications are time-invariant throughout the sample period: no state moved into or out of the unlimited or low-exemption category between 2011 and 2019, and the opt-in jurisdictions did not change their opt-in status. The binary extreme-exemption classification is therefore a fixed, time-invariant attribute of the state, not a consequence of the reform or of contemporaneous policy changes. The continuous

[9] New Jersey and Pennsylvania (zero statute) and Kentucky ($5,000 statute) opt in and so take the federal floor as their effective exemption; the other nine low-exemption sample states are opt-out, with statutes at or below roughly $25,000. The exact quarterly federal §522(d)(1) schedule is reported with the homestead-panel construction in the data section.

[10] In other unlimited-exemption states, the exemption protects against involuntary seizure but does not prohibit voluntary pledging. A borrower in Florida could, in principle, offer a second lien on the home as additional collateral. This attenuates the treatment in those states relative to Texas, where the constitutional prohibition makes the lender's exclusive dependence on business collateral exact.

effective-exemption measure does carry within-state variation, from the triennial federal floor common to all opt-in states and from a small number of state statutory changes during the window, so for the continuous design the time-invariance claim applies to the category, not to the exemption level.[11]

The empirical design exploits this variation in two forms. The primary specification uses the continuous effective exemption across all jurisdictions. The secondary specification restricts to seven unlimited- and twelve low-exemption states, sharpening the contrast between regimes where recovery rests almost entirely on business collateral and regimes where the home-equity buffer remains intact.

3 A Selection-and-Enforcement Framework

The introduction distinguished two kinds of private information the reform treats differently. This section formalizes the first, the lender's read on the recovery value of an asset class, as a model of how a lender rebuilds that read once standardization erases it; the relationship knowledge of a specific borrower is left aside because the reform never touches it. The model is deliberately stylized. Its job is to sign how a lender reshapes its collateral when valuation discretion is withdrawn, and so to organize the adaptation evidence that follows. Proofs and the micro-foundation of the selection cost are collected in the Internet Appendix. Table 1 collects the notation.

[11] The most pronounced within-state change is Ohio's homestead jump from roughly \$24,000 to \$125,000 in March 2013, well outside the unlimited-exemption group on either side; its robustness is examined in Section 6.3.

Table 1: Notation for the selection-and-enforcement framework.

Symbol	Meaning
Borrower primitives	
ϑ	Enforceability of a borrower's movable business collateral
γ_0	Correlation, across borrowers, between pledgeable real estate and movable-collateral enforceability
h	State homestead exemption
The standardized rule	
μ_m	Over-crediting of movable collateral (recovery shortfall per dollar of movable-collateral lending)
$\mu_{RE} \approx 0$	Real estate, left undistorted
Lender's choices	
s	Selection: tilt toward real-estate-collateralized loans (escapes the shortfall)
a	Enforcement: lien specificity on movable collateral (offsets the shortfall)
Parameters	
λ	Lender's retained pari-passu share of recovery
$m \equiv \lambda\mu_m$	Effective stake
ψ	Enforcement-cost parameter
$\phi(h) = h/\kappa$	Selection-cost curvature, rising in the exemption
κ	Density of homestead valuations
ϑ_0	Baseline movable enforceability (portfolio average at $s = 0$)
Optimum	
$s^*,\ a^*$	Optimal selection and enforcement

Environment. A lender finances a continuum of borrowers, each described by the real estate it can pledge and the enforceability ϑ of its movable business collateral, the fraction a lender recovers by perfecting specific liens. The two attributes are correlated across borrowers with sign γ_0: borrowers with more pledgeable real estate hold movable collateral that is more enforceable if $\gamma_0 > 0$ and less enforceable if $\gamma_0 < 0$. Standardizing valuation over-credits movable collateral, recognizing it at $(1 + \mu_m)$ times its conservative liquidation value with $\mu_m > 0$, while real estate, whose appraisal stays discretionary, is left undistorted ($\mu_{RE} \approx 0$). Because a borrower turned away re-shops, μ_m is a recovery shortfall the lender must make up on every dollar of movable-collateralized lending; it is the engine of the model, making the movable class a liability and real estate a refuge (Manove et al., 2001). The defensive margins are dormant when $\mu_m = 0$ and activated by the standardization, much as standardized capital rules lead banks to work the inputs they still control (Behn et al., 2022; Plosser and Santos, 2018).

The exemption prices selection, not recovery. A homestead exemption h shields the borrower's home equity from *involuntary* creditors, but a borrower who consents to a home lien waives that shield, so a lien the lender selects into recovers in full regardless of h (Berkowitz and White, 2004; Gropp et al., 1997). The exemption therefore does not scale recovery on a consented home lien; it scales the *price* of obtaining one. A borrower who values the shield more demands a larger inducement to pledge, so where the exemption is high willing pledgers are scarce and dear. This is a distinct role from the one the exemption plays in the pricing design, where it shields the home from *involuntary* deficiency claims and so determines how far recovery leans on business collateral; the two are consistent, since a higher exemption both makes consensual real-estate selection dearer here and concentrates involuntary recovery on business assets there. Aggregating these pledge-or-waive decisions (Internet Appendix) yields a convex selection cost $C(s;h) = \frac{1}{2}\phi(h)\, s^2$ whose curvature $\phi(h) = h/\kappa$ rises in the exemption, with κ indexing the density of homestead valuations. In Texas, where business-purpose home liens are barred, selection is unavailable at any price, the corner $h \to \infty$.

The lender's problem. Having lost valuation discretion, the lender defends along two fronts. *Selection* s tilts the portfolio toward real-estate-collateralized loans, an asset class the rule does not over-credit and on which a granted lien recovers in full, and so *escapes* the shortfall; tilting changes the average enforceability of the movable collateral that remains, $\vartheta_0 + \gamma_0 s$, through the borrower correlation. *Enforcement* a perfects more specific liens on the movable collateral the lender keeps and so *offsets* the shortfall, recovering $a(\vartheta_0 + \gamma_0 s)$ per unit of effort. The lender retains a uniform pari-passu share λ of every recovery; writing the effective stake $m \equiv \lambda\mu_m$, its gain over the conservatively valued benchmark is

$$\Pi(s,a) \;=\; m\big[\, s + a\,(\vartheta_0 + \gamma_0 s)\,\big] \;-\; \tfrac{1}{2}\,\phi(h)\,s^2 \;-\; \tfrac{1}{2}\,\psi\,a^2, \tag{1}$$

the value of escaping plus offsetting the shortfall, net of the convex selection cost and a convex enforcement cost in ψ. The payoff applies the enforcement view of secured debt, that a lien is worth what the lender can perfect and recover, to a valuation rule rather than a financing constraint (Rampini and Viswanathan, 2025; Inderst and Mueller, 2007); selection is the dual collateral-choice decision of which asset class to lend against (Donaldson et al., 2020; Calomiris et al., 2017). Because λ enters only through m, the guarantee scales the lender's effort but cannot tilt it toward either margin.

Comparative statics. The first-order conditions are $\phi(h)\,s^* = m(1+\gamma_0 a^*)$ and $\psi\,a^* = m(\vartheta_0 + \gamma_0 s^*)$, with a unique interior optimum under a bounded-complementarity condition (Internet Appendix). The exemption enters only through the selection-cost curvature $\phi(h)$, and differentiating the optimum delivers two signs.

Proposition 1 (Selection weakens with the exemption)**.** $\partial s^*/\partial h < 0$*: the lender tilts its portfolio less toward real estate where the exemption is higher.*

A higher exemption makes the homestead shield dearer to surrender, so the marginal pledger costs more and the escape route narrows. The sign holds for every value of γ_0; this is the model's robust prediction.

Proposition 2 (The enforcement gradient carries the correlation's sign)**.** $\text{sign}(\partial a^*/\partial h) = -\,\text{sign}(\gamma_0)$.

Corollary 1 (Diversification versus concentration)**.** *Because γ_0 is the correlation between the lender's two recovery sources, it governs whether they are diversified or concentrated. If $\gamma_0 < 0$, the sources are diversified: enforcement rises as selection falls and the lender substitutes one defense for the other across the exemption distribution. If $\gamma_0 > 0$, they are concentrated: both defenses weaken together, leaving the lender most exposed where the home is most shielded. The case $\gamma_0 = 0$ is the knife-edge, with selection falling and enforcement flat.*

As selection recedes the portfolio becomes more movable-collateralized, and whether the residual movable collateral is then more or less enforceable depends on which borrowers populate it. When real-estate borrowers hold *less* enforceable movable collateral ($\gamma_0 < 0$), the two recovery sources are negatively correlated and so diversify: the residual portfolio is better and enforcement rises as selection dies. When they hold *more* ($\gamma_0 > 0$), the sources are positively correlated and concentrated: the residual portfolio is worse and both defenses weaken together. The natural prior, that wealthier real-estate borrowers also have cleaner movable collateral, is the concentration case, so the diversification that lets enforcement substitute for selection requires the opposite correlation. Figure 1 traces the two regimes.

What the model delivers. The model's empirical content rests on Proposition 1, the always-signed prediction that collateral composition tilts off real estate where the exemption is high. Proposition 2 converts the sign of the specificity gradient into a reading of the borrower correlation; but because that reading is inferred from the gradient itself rather than from independent evidence on γ_0, and the filing data cover a single high-exemption state, I treat the regime it implies as an ex-post interpretation rather than a test. The model signs only how the lender reshapes collateral; it is silent on the level of the pricing gradient, which is identified by the homestead-moderated difference-in-differences design rather than derived from the model. The collateral evidence that follows speaks to both predictions.

4 Data

4.1 SBA 7(a) Loan Records

The primary dataset consists of administrative loan-level records from the SBA's 7(a) program, obtained via the Freedom of Information Act.[12] Each record contains the gross approval amount, interest rate (fixed or variable), loan term, guarantee percentage, approval and disbursement dates, borrower name, address, state, two-digit NAICS code, lender identity and location, and loan status. I restrict to loans approved during 2011Q1–2019Q4, providing a balanced pre- and post-reform window around the January 2014 collateral standardization.

Loan spreads are constructed using daily Treasury yields (3-month T-bill and 1-, 2-, 3-, 5-, 7-, 10-, 20-, and 30-year maturities) from FRED and the U.S. Treasury. For variable-rate loans, the spread equals the stated interest rate minus the 3-month T-bill rate on the approval date. For fixed-rate loans, the spread equals the stated rate minus the piecewise-linearly interpolated Treasury yield curve rate at the loan's maturity.

4.2 UCC Filings and Collateral Classification

UCC Article 9 governs security interests in *personal property* (that is, movable assets such as equipment, inventory, accounts receivable, chattel paper, and general intangibles).[13] A UCC filing does not, and legally cannot, perfect a security interest in real estate. This statutory scope makes UCC match status a natural proxy for collateral type: loans suc-

[12] Public data via the SBA Open Data portal (`data.sba.gov`); Brown and Earle (2017) analyze 7(a) effects on firm employment using similar data.

[13] UCC Article 9, §9-109(a)(1): "this article applies to a transaction, regardless of its form, that creates a security interest in personal property." Real property (land, buildings, fixtures affixed to land) falls outside Article 9 and is governed by state real estate law. The one exception is fixtures: goods that become attached to real property may be perfected under Article 9 (§9-334), though they are classified as movable property for UCC purposes.

cessfully matched to a UCC filing are backed primarily by movable business assets, while unmatched loans are more likely backed by real estate or a mixed collateral base that includes immovable property. The spread reduction should be largest where the fraction of expected recovery from business collateral is high, which corresponds to movable business assets. Working in the opposite direction, the post-reform preset, uniform schedule is at least weakly more generous than the lender's typical pre-reform discretionary appraisal, and the 0.85 preset on real estate is unambiguously generous, so matched loans need not show the larger reduction. The two forces push in opposite directions on the matched-versus-unmatched contrast, and which dominates is an empirical question resolved in the results.

Each filing records the debtor, the secured party, and a free-text collateral description detailing what assets the lender has a lien against. Two filing types anchor the analysis. A *floating* (or *blanket*) lien claims all of the borrower's business assets, including after-acquired property, so the lien attaches to a replaceable pool that shifts as the borrower buys and sells in the ordinary course. A *specific pledge*, by contrast, fixes on identified assets that the borrower cannot dispose of without the lender's consent. I locate each filing on the spectrum between these poles with three measures: a *blanket-scope indicator* (one if the filing claims all assets in generic language), a *scope score* (a weighted count of floating-lien clauses, after-acquired property, proceeds, and all-locations language), and a *specificity score* (a count of asset-specific identifiers such as serial and model numbers). The distinction matters because the two filing types carry different information requirements: a floating lien conveys little asset-specific information and therefore requires little, whereas a specific pledge obliges the lender to produce collateral-specific information about the identified assets. This collateral-level detail is unavailable in the SBA administrative data, which records only whether collateral was pledged, not its composition. Linking SBA loans to their UCC filings therefore enables direct observation of what the lender actually claimed as collateral, how broadly the lien was drafted, and how much detail the lender invested in specifying individual asset categories, the variation needed to test whether lenders rebuild the asset-class information the reform took away.

UCC filings are maintained at the state level and are not available through any centralized federal source. I obtain filings from three states: California and Florida, where I web-scrape the state filing office portals, and Colorado, where filings are publicly available through the state open data portal.[14] The restriction to three states reflects data availabil-

[14]Colorado UCC filings are published at `data.colorado.gov`. California and Florida filings are obtained through automated extraction from the respective Secretaries of State online filing systems. See the replication package for extraction scripts.

ity, not sample design: no centralized UCC database exists, and most states do not provide bulk electronic access to filing records. These three states account for approximately 20% of SBA 7(a) lending volume and span diverse economic environments (California's technology and agriculture sectors, Florida's services and tourism, Colorado's energy and professional services), providing reasonable coverage despite the constraint. Within the UCC sample, identifying variation for the binary moderator design comes from Florida (the only unlimited-exemption state with UCC coverage); California and Colorado contribute to the continuous specification, where their effective exemptions lie in the interior of the distribution.

I link SBA loans to UCC filings through an industrial-scale fuzzy matching pipeline that combines name-based blocking, geographic verification, and a restricted time window around the loan approval date. Each candidate pair receives a composite score incorporating name similarity and geographic proximity, with tiered acceptance gates that balance precision against recall. Match quality is evaluated through repeated manual verification of random samples across all three states; precision exceeds 95%, with the majority of retained matches classified as high confidence. The estimation sample for the collateral composition analysis comprises 18,341 classified loans across California, Florida, and Colorado.

Collateral classification. Each matched filing's collateral description is classified into one of 14 Article 9 asset categories. Blanket liens (filings that claim "all assets" of the debtor) dominate the regression sample (approximately 80% of >$350K loans), consistent with the SBA's collateral adequacy requirement. Within blanket liens, I distinguish generic filings ("all assets of debtor") from itemized filings that specify individual asset categories, a distinction that captures the lender's information acquisition.

Filing intensity index and lender-level intensity change. The composite filing intensity index standardizes three measures within the blanket lien subsample: an inverse blanket-scope indicator (rewarding itemized over generic filings), a specificity score (asset-specific identifiers normalized by filing length), and the number of distinct Article 9 categories covered. Higher values reflect more targeted information acquisition.

To illustrate the variation these metrics capture, consider two filings drawn at random from the matched Florida sample, one generic and one itemized.[15]

[15]The two filings are drawn at random from the matched Florida sample by `scripts/R/18_ucc_example_filings.R` (seed 20140101), which samples one blanket-scope filing from the pre-reform period and one non-blanket filing from the post-reform period. The quoted text is verbatim from the public financing statement, anonymized only by masking the manufacturer model and serial

Generic (Florida, ≈2014, blanket scope): "This financing statement covers the following collateral: All assets."

Itemized (Florida, ≈2016, non-blanket): "This FINANCING STATEMENT covers the following collateral: ONE (1) NEW 2016 [Manufacturer Model] VIN #XXX 26′ REEF BODY WITH [Manufacturer Model]."

These are verbatim (anonymized) filings, terser than a constructed illustration and representative of the language lenders actually file. The generic filing establishes a nominal legal claim with no asset-specific information; "All assets" reasonably identifies the collateral under the public-notice standard but reveals nothing about the borrower's asset base. The itemized filing identifies the condition ("NEW"), year, make, model, body type, and serial number of specific equipment, revealing that the lender examined the borrower's collateral in detail. In the filing intensity index, the generic filing scores low on all three components, while the itemized filing scores high on specificity (asset-level identifiers).

The filing intensity index measures the lender's investment in publicly observable collateral specificity on the UCC financing statement: the blanket scope and specificity of the collateral description on the public notice filing. It does not measure enforceable collateral scope (which depends on the unobserved security agreement), private valuation knowledge (which the lender may hold without encoding in the filing), or the content of the security agreement itself.[16] The mapping from observable filing intensity to the lender's underlying private read on business-asset recovery value is assumed monotone but cannot be validated with filing data alone. I treat this as a maintained assumption, and the empirical interaction is read as a descriptive decomposition rather than a structurally identified object.

The *filing-intensity change* variable measures each lender's pre-to-post shift in mean filing intensity index within the blanket lien subsample, for lenders with filings in both periods. Exploiting the *change* rather than the level sidesteps the fact that a generic pre-reform

digits.

[16]Two documents govern a secured transaction. The *financing statement* (UCC-1) is a public notice filing governed by UCC §9-504, which requires only that the filing "reasonably identifies" the collateral; a filing that says "all assets" satisfies this standard. The *security agreement* is a private contract governed by §9-108(c), which requires a description "sufficient to identify what is described"; a security agreement that says "all assets" is unenforceable. The financing statement perfects the lien; the security agreement creates it. I observe only the financing statement. The gap between the two documents is one-directional: lenders routinely overfile, claiming "all assets" on the UCC-1 while the underlying security agreement covers a narrower set, because the $20 filing cost is trivial relative to its option value of blocking subsequent creditors. Itemized filings are less ambiguous, because a 200-word collateral description with serial numbers and model years is almost certainly drawn from the same collateral schedule used in the loan documents. Neither source of error threatens identification because the paper's estimand is the *differential change* in filing behavior across homestead exemption regimes, not the level of collateral coverage; overfiling and template conventions are plausibly time-invariant within lender and absorbed by lender fixed effects.

filing is uninformative about collateral composition: a lender who shifts from generic to itemized filings at the reform boundary reveals a behavioral response to the policy change even if the pre-reform filing was uninformative. This variable serves as the moderator in the test of whether lenders rebuild asset-class information where home equity offers no recovery buffer: the triple interaction of treatment, post, and the filing-intensity change is interpretable as a conservative test of attenuation in asset-specific information acquisition under the maintained assumption that filing noise is orthogonal to the treatment-by-post interaction conditional on lender and quarter fixed effects, plausible because these filing conventions are time-invariant within lender and orthogonal to the exemption regime (preceding footnote).

4.3 Homestead Exemption Panel

State-level homestead exemption limits are constructed from the panel compiled by Indarte (2023), which I verify against primary state statutes and extend to cover the full sample period.[17] For the continuous treatment specification, I use the *effective* homestead exemption, defined as the maximum of the state statutory amount and the federal §522(d)(1) amount when the state permits debtors to elect the federal exemption.[18] For the binary design, I restrict to the seven unlimited-exemption states (AR, FL, IA, KS, OK, SD, TX) and twelve low-exemption states (effective exemption $\leq$ \$23K for most of the sample window: AL, GA, IL, IN, KY, MD, MO, NJ, PA, TN, VA, WY). All twelve states have effective exemptions at or below the federal floor, which protects less than a quarter of typical home equity for SBA borrowers in this size tier, leaving the lender with substantial recovery through deficiency judgment. Texas is the corner case within the unlimited-exemption group: business-purpose home-equity liens are constitutionally barred (Section 2.4), so accessible home equity is exactly zero in Texas, while the other six unlimited-exemption states (AR, FL, IA, KS, OK, SD) protect against involuntary seizure but permit voluntary pledging. No state moves between the unlimited-exemption and low-exemption groups during 2011Q1–2019Q4, so the binary indicator Treated_s is a fixed attribute of the state; the

[17]The base panel is publicly available at `sashaindarte.github.io/public_goods/`. I verify each state's exemption amount against the relevant statute and extend the panel to include monthly observations through 2019, county-level variation for California, New York, and Washington, and the federal bankruptcy exemption option for states that permit it.

[18]Twenty states plus DC are "opt-in" jurisdictions that permit debtors to elect the federal bankruptcy exemption in lieu of the state amount (11 USC §522(b)(2)). Among the twelve low-exemption states in the binary design, three are opt-in (Kentucky, New Jersey, Pennsylvania); these states have statutory amounts below the federal floor and are therefore coded at the federal amount. The federal §522(d)(1) amount during the sample period is \$21,625 (2011Q1–2013Q1), \$22,975 (2013Q2–2016Q1), \$23,675 (2016Q2–2019Q1), and \$25,150 (2019Q2–2019Q4).

continuous effective-exemption measure varies within state only through the triennial federal §522(d)(1) updates, which are common to all opt-in states and absorbed by quarter fixed effects. Results are robust to modest shifts in the threshold.

Lenders are classified as banks (matched to FDIC via institution number), credit unions (via NCUA number), or non-bank lenders (neither). Multi-state lenders are identified as those operating in both unlimited-exemption and low-exemption states within the sample.

4.4 Key Variables

The binary treatment indicator $\text{Treated}_s = 1$ for the seven unlimited-exemption states and 0 for the twelve low-exemption states. $\text{Post}_t = 1$ for loans approved on or after January 1, 2014. For the continuous specification, I use $\log(\text{effective exemption}_{s,t})$ and a rank-based measure across all jurisdictions.[19] Controls include log loan amount, term (months), a fixed-rate indicator, a relationship lending indicator (equal to one if the borrower has a prior SBA loan from the same lender), a previous-SBA indicator, a collateral indicator, a corporation indicator, a franchise indicator, and a revolver indicator.

Borrower identities are constructed via union-find fuzzy record linkage on standardized names within state. Repeat borrowers are those with at least two SBA loans within the \$350K–\$1M tier; post-entry status is defined by the first loan's approval date. The indicator PreviousSBA_i equals one if the borrower has any prior SBA loan from any lender, serving as the empirical proxy for the borrower-relationship information that the reform never touches: within-borrower spreads should be unchanged, and the incumbent lender's hold on a repeat borrower should, directionally, be stronger where a larger share of the loan is backed by business collateral.

Collateral composition outcomes. Five variables measure collateral coverage. The *blanket lien indicator* equals one if the filing claims all business assets. The *scope score* (0–5) is a weighted count of floating-lien clauses: after-acquired property, proceeds, and all-locations language. The *blanket-scope indicator*, defined within the blanket lien subsample, equals one if the filing uses generic "all assets" language without itemizing specific categories; a decrease means fewer filings use generic language. The *specificity score*, also defined within the blanket lien subsample, counts asset-specific identifiers (serial num-

[19] Unlimited-exemption states are coded at \$5,000,000 following Indarte (2023), yielding $\log(5{,}000{,}000) \approx 15.42$. The minimum effective exemption across the 49 jurisdictions is \$5,000, so log(effective exemption) is well-defined everywhere without a $+1$ offset. The rank specification (column 4 of Table 6) assigns unlimited-exemption states to rank 1.0, providing a coding-independent robustness check.

bers, make, model, year) normalized by filing length. The *composite filing intensity index* is described in the UCC data subsection above.

Falsification outcomes. Falsification tests use state-quarter log loan counts and lending volume, the repeat-borrower share, charge-off rates, and Express-program and pre-reform placebos. Results are reported in the robustness subsection.

4.5 Sample Construction and Summary Statistics

The full SBA 7(a) dataset contains approximately 117,000 Standard Guaranty loans above \$350K approved during 2011Q1–2019Q4. The \$350K threshold is economically motivated: it fixes a uniform guarantee rate ($g = 0.75$) and a uniform collateral adequacy standard (the "fully secured" requirement with preset haircuts), and excludes two contemporaneous smaller-loan policy changes (detailed in Section 5.1).

Data cleaning. I drop loans in U.S. territories (GU, MP, VI, PR, AS) and geographically non-contiguous states (AK, HI), which face distinct lending environments and housing markets. Spreads are winsorized at the 1st and 99th percentiles to limit the influence of extreme values. To ensure that lender fixed effects are estimated from a sufficient number of observations, I retain only lenders with at least six loans per period (pre- and post-reform), yielding 409 lenders accounting for 86$\%$ of loans in the estimation sample. The filter conditions on post-reform survival, which could in principle absorb lenders whose post-reform activity was differentially affected by the reform; the falsification analysis shows that the log number of distinct lenders per state-quarter is essentially unchanged at the reform boundary, in both the binary and continuous specifications, with effects close to zero and statistically insignificant (untabulated), ruling out differential lender exit as a confound.

Estimation samples. After applying the cleaning protocol, the primary spread analysis uses the full-state sample (103,879 loans, 103,506 after singleton removal) with the continuous exemption measure. For the binary comparison, I restrict to nineteen states (seven unlimited-exemption, twelve low-exemption with exemption $\leq$ \$23K), yielding 44,034 loans (43,770 after singletons). For the within-borrower spread test (the borrower-fixed-effects specification), I identify repeat borrowers within the \$350K–\$1M tier who obtained at least two SBA loans during the sample period. The all-states repeat-borrower panel contains 3,210 borrower pairs (3,131 after singletons); within the nineteen-state binary sample, 1,153 pairs (911 after singletons). The small size of this panel reflects the relative rarity of repeat SBA borrowing: approximately 4.7% of first-loan borrowers in the \$350K–\$1M tier obtain a second loan within the sample window, with similar rates in treated and control states.

For the collateral composition analysis (Design 2), the UCC-matched sample covers 18,341 loans across California, Florida, and Colorado. The all-states sample ($N = 103{,}506$ after singleton removal) is used for the continuous treatment specification.

Summary statistics. Table 3 reports summary statistics for the full-state sample, split at the median homestead exemption. Panel A describes the loan-level sample (103,879 loans across 48 contiguous states and the District of Columbia). Mean loan amounts rise from approximately \$1.11 million pre-reform to \$1.23 million post-reform in both high- and low-HE groups, with similar term lengths (approximately 206–214 months). These are substantial loans to established businesses, not microenterprise credit; the typical borrower is an incorporated entity (87–95% corporation share) financing a major capital expenditure. Pre-reform spreads average 4.93 pp in high-HE states and 5.12 pp in low-HE states, a 19-basis-point level difference that is absorbed by state and lender fixed effects in all specifications; identification relies on parallel trends in spreads around the reform, not level equality. Both groups experience spread compression post-reform, but the compression is differential, the pattern examined in the event study. Collateral rates are high in both groups (88–93%), confirming that the "fully secured" requirement binds: nearly all loans in this size tier are collateralized, so the reform's haircut schedule applies to the vast majority of the sample. The homestead exemption row confirms the sharp treatment contrast: high-HE states average \$1.6–1.7 million in exemption (driven by unlimited-exemption states), while low-HE states average approximately \$23K.

Panel B describes the repeat-borrower subsample (3,210 pairs, all states). First-loan characteristics are broadly similar across groups: mean loan amounts of approximately \$610K, pre-reform spreads of 5.0–5.2 pp (4.8–4.9 pp post-reform), and term lengths of 185–196 months. The pre-reform first-loan spread is 17 basis points lower in high-HE states than in low-HE states, a level difference that does not threaten identification because the DiD relies on parallel *trends* around the reform, validated by the formal event study below. Covariates that differ in levels (corporation share, collateral share) are included as controls in all specifications. Appendix Table IA.4 reports the corresponding statistics for the sharp binary design (seven unlimited- versus twelve low-exemption states), and Appendix Table 4 provides a formal comparison of pre-reform borrower characteristics.

Figure 2 plots mean spreads for the binary subsample (seven unlimited- versus twelve low-exemption states) over the sample period. Both groups decline from approximately 5.2–5.4 pp in 2011 to 4.9–5.1 pp in 2013, moving in parallel before the reform. After January 2014, a visible gap emerges: spreads in unlimited-exemption states fall more steeply, consistent with the reform reducing pricing most where business collateral dominates the lender's recovery. The formal event study confirms that pre-reform coefficients are jointly insignificant under the preferred NAICS2 $\times$ quarter specification, so the pre-reform trends are statistically parallel; Table IA.6 reports both the preferred test and its additive counterpart. Inference is conducted with standard errors clustered at the lender level (409 clusters in the full-state sample); the robustness subsection reports additional checks.

5 Empirical Strategy

The primary analysis uses a difference-in-differences comparing Standard Guaranty loans above $350K across homestead exemption regimes, before and after the January 2014 collateral standardization. This design estimates the differential effect of the reform on loan pricing, with heterogeneity tests by loan size and collateral type. The state homestead exemption grades exposure to the reform: where the exemption is large, recovery in default concentrates on the business collateral whose valuation the reform standardized, so its pricing bite is largest there. A secondary design examines collateral composition and information acquisition using UCC filing data, providing direct evidence on how lenders adapted. The Linking subsection below explains how the two designs connect despite using different control groups and different samples.

Three expectations, each tied to a piece of the identifying variation, organize the analysis. First, the reform should lower loan spreads most in states where business collateral is the lender's main recovery channel, so the more a state's homestead exemption shields the borrower's home and other personal assets, the larger the spread reduction. Second, for a borrower who stays with the same lender across the reform, the spread should not change, so the exemption gradient should be a cross-sectional new-borrower phenomenon rather than within-borrower repricing. Third, where lenders cannot fall back on home equity, they should rebuild the asset-class information the reform took away by filing more detailed, asset-specific collateral descriptions, while where home equity provides a recovery buffer, lenders should instead file generic blanket liens. Designs 1 and 2 below take these in turn.

5.1 Design 1: Homestead Exemption DiD

Treatment variation. The collateral reform replaced discretionary valuations with a preset, uniform haircut schedule for all SBA 7(a) loans. The reform's impact on spreads depends on how much the lender's recovery relies on business collateral, which varies continuously with the state homestead exemption: where the home and other personal assets are sheltered from a deficiency claim, recovery leans almost entirely on business collateral, so the lender's pre-reform discretionary appraisal of those assets carried more pricing weight. Standardizing that appraisal removes the lender's private valuation advantage (and the preset, uniform schedule removes the compliance risk that rode on it), and that matters most where recovery concentrated on business assets. The spread gradient follows: borrowers in high-exemption states are more sensitive to the reform because business collateral constitutes a larger share of the lender's recovery.

The estimand is the differential effect of collateral standardization on loans in high-exemption states relative to low-exemption states: under the continuous specification, the average marginal effect of exemption intensity; under the binary specification, the average treatment effect on the treated (ATT) for unlimited-exemption states. The primary specification exploits this continuous variation across all states using $\log(\text{effective exemption}_{s,t})$ and rank-based measures, interacted with a post-reform indicator. This continuous treatment design uses all Standard Guaranty loans above \$350K across 48 contiguous states and the District of Columbia ($N \approx 104{,}000$), does not depend on any binary classification, and identifies the gradient from the full exemption distribution. For interpretability, I also restrict to an extreme binary comparison between seven unlimited-exemption states (AR, FL, IA, KS, OK, SD, TX; $\text{Treated}_s = 1$) and twelve low-exemption states (exemption $\leq$ \$23K; $\text{Treated}_s = 0$), which produces the cleanest coefficients for the spread gradient.

Sample restriction. I restrict to Standard Guaranty loans above \$350K (the collateral indicator is retained as a loan-level control; see Section 4). The \$350K threshold ensures a uniform guarantee rate ($g = 0.75$), a uniform collateral adequacy requirement ("fully secured" under the preset haircut schedule), and eliminates two contemporaneous policy changes affecting smaller loans (the October 2013 fee waiver for loans $\leq$\$150K and the July 2014 credit scoring mandate for loans $\leq$\$350K). The sample period is 2011Q1–2019Q4, providing a balanced pre- and post-reform window.

Spread specification. The primary specification exploits the continuous exemption variation across all states:

$$\text{Spread}_{ilst} = \beta \left(\log(\text{effective exemption}_{s,t}) \times \text{Post}_t\right) + X'_{ilst}\phi + \gamma_c + \mu_l + \tau_t + \epsilon_{ilst}, \quad (2)$$

where $\log(\text{effective exemption}_{s,t})$ is the log dollar homestead exemption in state s, Post_t equals one for loans approved on or after January 1, 2014, X_{ilst} is a vector of loan-level controls (log amount, term, fixed-rate indicator, relationship lending, previous SBA experience, collateral indicator, corporation, franchise, and revolver indicators), γ_c denotes county fixed effects, μ_l denotes lender fixed effects, and τ_t denotes year-quarter fixed effects. The coefficient of interest is β: the marginal effect of exemption intensity on the post-reform spread change, which the spread-gradient logic predicts is negative. For the binary comparison (Appendix Table IA.5), $\text{Treated}_s \in \{0, 1\}$ replaces $\log(\text{effective exemption}_{s,t})$ in (2), where $\text{Treated}_s = 1$ for the seven unlimited-exemption states and county fixed effects absorb the state-level main effect. The preferred specification replaces additive time effects with NAICS2 $\times$ quarter interactions (see the fixed effects strategy below).

Within-borrower spread specification. To separate cross-sectional repricing from within-relationship repricing, I add borrower fixed effects to the spread DiD and estimate it on the repeat-borrower panel: borrowers who obtained at least two SBA loans during the sample period, so that pre- and post-reform pricing can be compared within the same borrower (see Section 4). The specification is

$$\text{Spread}_{ilt} = \beta \left(\log(\text{effective exemption}_{s,t}) \times \text{Post}_t\right) + X'_{ilt}\phi + \xi_b + \mu_l + \tau_t + \nu_j + \epsilon_{ilt}, \quad (3)$$

where ξ_b denotes borrower fixed effects, μ_l denotes lender fixed effects, τ_t denotes quarter fixed effects, ν_j denotes two-digit NAICS industry fixed effects, and X_{ilt} includes loan-level controls (term, fixed-rate indicator, collateral indicator, corporation, franchise, and subprogram indicators). The borrower fixed effect absorbs all time-invariant borrower characteristics including state of operation; the interaction $\log(\text{effective exemption}) \times \text{Post}$ is identified from within-borrower variation in Post across the homestead exemption gradient. The prediction is that within-borrower spreads are unchanged: an existing borrower's price is disciplined by what a competitor would offer, and the competitor prices off pooled borrower-type information the reform does not touch, so the gradient is a cross-sectional new-borrower phenomenon rather than within-borrower repricing. Because the cross-sectional gradient in (2) is already identified across borrowers, this within-borrower specification serves as a consistency check: a β in (3) that shows no detectable within-borrower movement, set against the negative cross-sectional gradient, is consistent with locating the reform's pricing effect at the formation of new borrower-lender matches. The repeat-borrower panel is small, so this test has limited power; it corroborates the cross-sectional reading rather than establishing within-borrower invariance on its own. Standard errors are clustered at the lender level.

Heterogeneity tests. A loan-size heterogeneity test estimates the binary Treated $\times$ Post version of (2) on the seven-versus-twelve state sample, separately by loan-size bin ($350K–$1M, $1M–$2M, above $2M), to assess whether pricing effects vary with the share of the loan backed by business collateral.

Identification. Identification hinges on the parallel trends assumption: absent the reform, spreads would have followed the same trajectory across exemption regimes. I test this with an event study for the binary unlimited-versus-low-exemption contrast that plots year-by-year Treated $\times$ Year coefficients, with 2013 as the reference period; this binary contrast is the sharpest visual test of the gradient that drives the continuous specification. A joint test of the pre-reform coefficients does not reject the null of parallel pretrends at conventional levels under the preferred NAICS2 $\times$ quarter specification, though the 2012 coefficient is individually positive and marginally significant (Table IA.6 reports the joint test and the additive counterpart). Because any residual pre-trend in 2012 is positive while the treatment effect is negative, it works against finding the reduction I estimate. The SBA announced the reform on September 25, 2013, with a January 1, 2014 effective date, leaving roughly one quarter between announcement and implementation.[20] As a robustness check, I exclude the transition quarters (Q4 2013 and Q1 2014) to guard against anticipatory adjustment over that window. As a further robustness check, I restrict to 2011Q1–2016Q4 to guard against the 2015 Fed rate-hike cycle interacting differentially with homestead exemption regimes.

Identification also requires that a borrower's potential spread depend only on its own state's exemption regime, not on the reform's effect in other states. Because many lenders operate across treatment and control states, this rules out cross-border repricing in which a lender adjusts terms on its low-exemption-state loans in response to its treated-state exposure; the lender fixed effects absorb time-invariant lender pricing levels, and the NAICS2 $\times$ quarter effects absorb common shocks, leaving such contamination as the residual threat. Two further concerns bear on identification. First, outside Texas, borrowers in

[20]The reform was issued through SBA Information Notice 5000-1290, "Issuance of SOP 50 10 5(F)," dated September 25, 2013; the printed effective date on the SOP 50 10 5(F) cover page is January 1, 2014.

unlimited-exemption states may voluntarily pledge home equity, attenuating the effective treatment. Texas is the only state where a deficiency claim cannot reach the home at all (business-purpose home liens are constitutionally barred; see Section 2.4), so the Texas-only binary DiD provides a clean upper bound while the continuous specification estimates the gradient without relying on any binary classification. Second, the binary design uses nineteen states (seven treated, twelve control), raising small-cluster inference concerns. The continuous specification across 49 jurisdictions (48 contiguous states and the District of Columbia), wild cluster bootstrap (MacKinnon et al., 2023), and the falsification tests together address these concerns.

5.2 Design 2: Collateral Specificity

The reform should change both the information acquisition and the composition of pledged collateral: standardizing valuation strips the lender's private read on business-asset recovery value, so a lender whose recovery depends on those assets rebuilds it through precise itemized filings, while a lender with a home-equity cushion takes the cheapest route to perfection, a generic blanket lien. In the language of Section 3, collateral composition traces the selection margin and filing specificity traces the enforcement margin. Because UCC filings record only movable (Article 9) collateral, selection is observed through its complement, the shift between movable and real-estate backing, rather than directly, which is why the real-estate substitution is the less precisely identified of the two. I test this directly using UCC filing data matched to SBA loans across California, Florida, and Colorado (18,341 classified loans in the estimation sample). UCC filings record security interests in personal (movable) property under Article 9 of the Uniform Commercial Code; the filing descriptions reveal what assets the lender perfected a lien against and how much the lender invested in information acquisition about those assets.

I estimate a general DiD specification across multiple collateral outcomes y_{ilst}:

$$y_{ilst} = \beta\,(\text{Treated}_s \times \text{Post}_t) + X'_{ilst}\phi + \mu_l + \tau_t + \nu_j + \gamma_c + \epsilon_{ilst}, \tag{4}$$

where $\text{Treated}_s = 1$ for Florida (unlimited homestead exemption) and 0 for California and Colorado (limited exemption, $\sim\$75\text{K}$), ν_j denotes two-digit NAICS industry fixed effects, and γ_c denotes county fixed effects.[21] I cluster standard errors at the lender level; the three-state UCC composition sample contains 211 distinct lenders, of which 185 enter the clustered variance estimate as effective clusters after the fixed effects absorb single-

[21] A continuous specification using $\log(\text{effective exemption}_{s,t})$ across all three states is reported as a robustness check.

ton lender-cells. Because the reform itself raises the filing rate, conditioning the composition and specificity outcomes on the matched subsample selects on a treatment-affected variable, so the gradient could in part reflect differential selection into the matched sample rather than within-lender intensification. The specificity gradient is robust to inverse-match-probability weighting and to within-collateral-type fixed effects (Table IA.2); these address selection on observable characteristics into the matched sample. Selection on unobservables among the marginally matched loans cannot be fully ruled out, and the specificity gradient rests on a single treated cluster, Florida, so a Florida-specific filing shock cannot be excluded.

I examine three sets of outcomes. First, *collateral composition*: the blanket lien probability, the scope score (a weighted count of floating-lien clauses including after-acquired property and proceeds language), and asset-type indicators (equipment, fixtures). The shift toward generic blanket liens should be smaller in unlimited-HE states, where the home and other personal assets are out of reach in a deficiency claim, business assets are the primary recovery channel, and lenders retain an incentive to maintain targeted coverage. The scope score should increase more in unlimited-HE states as lenders invest in precise, asset-specific filings on their primary recovery channel.

Second, *information acquisition*. Within the blanket lien subsample, I distinguish generic filings from itemized filings and measure filing intensity using an inverse blanket scope indicator, a specificity score, and a composite filing intensity index (see the data section for construction and interpretation). Conditioning on blanket liens holds legal scope constant and isolates an information-acquisition margin: because all blanket filings perfect the same security interest under UCC §9-504, variation in specificity proxies for the lender's underlying due diligence rather than legal coverage. This is the within-blanket version of the same prediction: unlimited-HE lenders maintain higher asset-specific information acquisition post-reform because the home cannot be reached in a deficiency claim, leaving business collateral as the primary recovery channel, while limited-HE lenders can lean on the home-equity buffer and reduce filing intensity. The extensive-margin sign (whether a lender broadens at all) is firm; the form the broadening takes carries a revealed-preference reading, and additive lender fixed effects absorb the reform-invariant lender heterogeneity in filing conventions.

Third, *information-cost pass-through.* I estimate the triple interaction

$$\text{Treated}_s \times \text{Post}_t \times \text{EffortChange}_l,$$

where EffortChange_l is the pre-to-post change in lender-level filing intensity within blanket liens. I treat this as a descriptive decomposition connecting the collateral composition evidence to pricing outcomes, not as a test. The cross-sectional gradient between spreads and filing intensity has no clear sign a priori: the net spread response depends on whether rebuilding asset-specific information recoups the discretionary valuation rent the reform compressed (dampening pre-reform spreads through a reduced standalone pass-through) or instead raises the floor on collateral recognition (amplifying spread reductions). The pass-through decomposition is therefore descriptive, not a causal mediation estimate.

5.3 Linking the Designs

Designs 1 and 2 examine different margins of the same reform using different samples and control groups. The continuous specification $\log(\text{effective exemption}_{s,t})$ bridges across both by estimating the gradient from the full exemption distribution without relying on any binary grouping.

Different control groups by necessity. Design 1 uses low-exemption states (HE $\leq$ \$23K) as the benchmark; Design 2 compares Florida (treated) against California and Colorado (control), the three states where I have classified UCC filing data. This difference reflects data availability, not a design choice. The continuous $\log(\text{effective exemption})$ specification bridges the two designs by removing any dependence on binary treatment classification: Design 1 estimates the gradient on the 49-jurisdiction sample for spreads, together with the within-borrower spread test that places that gradient in the cross-section of new matches, while Design 2 estimates the analogous continuous specification for the collateral-adaptation outcomes (blanket lien share, scope score, filing intensity index) on the three-state UCC subsample, where alone these outcomes are observed. The FL-versus-CA+CO spread DiD within the UCC subsample is treated as a separate information-cost pass-through decomposition.

Information-cost pass-through as a descriptive decomposition. Design 1 estimates the average pricing effect; Design 2 measures the information acquisition response and decomposes how it co-moves with pricing through the information-cost pass-through triple interaction. This pass-through is a descriptive decomposition, not a causal mediation estimate, and the FL-versus-CA+CO comparison underlying it was chosen for data availability rather than as an independent test. The NAICS2 $\times$ quarter and county fixed effects absorb industry-specific and local trends, and the pre-reform trends for the collateral-adaptation outcomes are statistically parallel (Figures IA.2 and 5). The continuous specification within the three-state sample bridges to the all-states design without depending on Florida-specific shocks.

5.4 Inference

Standard errors are clustered at the lender level throughout the spread and collateral-composition specifications, reflecting lender-level variation in collateral valuation practices and pricing. The within-borrower spread specification (3) likewise clusters at the lender level. Routine inference robustness checks (wild cluster bootstrap of MacKinnon et al. (2023); county-clustered and two-way standard errors) are reported in the Internet Appendix (Tables IA.12–IA.13).

5.5 Fixed Effects Strategy

For spreads (Design 1), treatment varies at the state level, making industry-specific time shocks the primary confound. Oklahoma and Texas account for a large share of the treated group, and the oil price collapse (late 2014 through 2016) coincides with the post-reform window. The spread specification therefore uses NAICS2 $\times$ quarter interactions to absorb industry-specific business-cycle shocks while preserving the cross-state identifying variation. Lender $\times$ time interactions would restrict identification to multi-state lenders operating in both treatment groups simultaneously, a small and unrepresentative subset (fewer than a third of estimation-sample lenders). These are predominantly large national banks whose soft-information production differs from the modal SBA lender in the relationship-lending literature (Berger and Udell, 1995; Petersen and Rajan, 1994); the attenuation under this specification is therefore consistent with a reduction in the identified rent margin on this subset rather than evidence of confounding.

For spreads, any oil-related confound would bias the estimate toward zero (higher credit risk in oil states would raise, not lower, spreads), making the negative spread finding conservative.

Table 5 defines the empirical variables and proxies used throughout the analysis.

For collateral composition (Design 2), the nature of the confound is different. Collateral filing is an internal lender choice, not disciplined by competition: two lenders in the same state and quarter can have different filing conventions due to template updates, new legal counsel, or system migrations unrelated to the reform. Time-varying lender heterogeneity in filing practices is a concern, but the available data do not support a fully saturated lender $\times$ quarter specification within the three-state UCC subsample. I therefore report results under additive lender, quarter, two-digit NAICS industry, and county fixed effects, matching Equation (4), which absorb time-invariant lender heterogeneity, common time shocks, industry composition, and local conditions. The coefficients are stable across the controls included, consistent with the collateral adaptation pattern reflecting the reform rather than filing drift.

6 Results: Pricing

6.1 Average Spread Reduction

Using all Standard Guaranty loans above $350K across all jurisdictions (Table 6, column 1), the continuous DiD estimates a spread coefficient of -1.47 bp per unit of log effective exemption, significant at the 5% level: a 10 percent rise in the exemption lowers spreads by approximately 0.14 bp. Cumulating along the empirical range, the move from the minimum effective exemption of $5,000 to a mid-exemption $100,000 reduces spreads by approximately 4.4 bp, and the move from minimum to an unlimited-exemption state (coded at $5,000,000) yields approximately 10.2 bp, consistent with the binary 10.7 bp in Table IA.5. A complementary rank specification (column 4), scaling each state's percentile in the effective-exemption distribution to $[0, 1]$, yields a reduction of about 9.3 bp from bottom to top of the distribution, significant at the 5% level.

Spreads fall more in states where home equity is more protected and business collateral is the lender's primary recovery channel. The more a state's homestead exemption shields the borrower's home and other personal assets from a deficiency claim, the more the fraction of expected recovery concentrates on business collateral, and the more pricing weight the lender's pre-reform discretionary appraisal carried. Standardizing that appraisal removes the private valuation advantage, and the preset, uniform schedule removes the compliance risk that rode on it, so the spread reduction is largest precisely where recovery concentrated on business assets. The gradient holds across the full exemption distribution (Figure 3).

Restricting to the extreme HE regimes (seven unlimited-exemption versus twelve low-exemption states) produces coefficients that map directly to the homestead-exemption channel. The preferred specification, whose NAICS2 $\times$ quarter fixed effects absorb industry-specific business cycle shocks, including the oil price collapse that disproportionately affected Texas and Oklahoma, estimates a spread reduction of 10.7 bp, significant at the 1% level (Table IA.5, column 1). The additive-FE robustness column yields a virtually identical 10.6 bp (column 2), and adding a county-level home price control raises the estimate to 11.0 bp (column 3), indicating that neither differential house price trends nor industry composition drives the result. Because the binary design rests on nineteen states, I confirm the estimate with the wild cluster bootstrap, which rejects the null at conventional levels (Table IA.12).

The spread event study (Figure 4) is consistent with parallel pre-trends under the preferred NAICS2 $\times$ quarter specification, where the joint Wald test does not reject parallel pre-reform trajectories (Table IA.6); the only individually marginal pre-reform coefficient is in 2012, a residual marginal pre-reform difference that survives the NAICS2 $\times$ quarter effects but is positive while the treatment effect is negative, so it works against the estimated reduction. Post-reform, coefficients become negative and grow in magnitude, consistent with portfolio rotation from pre- to post-reform pricing as the reform applies only to new approvals.[22]

For the median treated-state loan ($1.17 million), the 10.7 bp spread reduction implies approximately $1,250 in annual interest savings. Across the approximately 15,100 treated-state loans in the post-reform sample, the aggregate annual reduction is approximately $19 million. The modest per-loan magnitude is what one expects when the 75% guarantee absorbs most of any collateral revaluation, limiting the lender's exposure to 25% of any valuation change.

The remainder investigates what drives this average: loan-size heterogeneity, lender adaptation on both the extensive and intensive margins, and the information-cost pass-through connecting filing behavior to loan pricing.

6.2 Loan-Size Heterogeneity

If the spread sensitivity to collateral valuation changes scales with the share of the loan backed by business collateral, the effect should be larger for smaller loans, where business-asset valuations represent a larger share of the collateral adequacy calculation. Table 7 reports the 7v12 binary DiD estimated separately by loan-size range.

The effect is stable across loan-size bins, ranging from about 11 bp for $350K–$1M loans to about 9–10 bp in the larger size ranges (Table 7); the estimate is significant at the 1% level in the smallest bin and marginal in the two upper bins. The estimates are statistically indistinguishable across loan-size bins, consistent with the reform operating through the homestead-exemption channel rather than a strong business-collateral-share gradient within the $350K+ tier.

[22]The reform was announced in September 2013 and effective January 2014. With 2013 as the reference year, the pre-announcement (2011) coefficient is modestly positive and individually insignificant, and the joint Wald test fails to reject parallel pre-reform trajectories (Table IA.6). Post-reform coefficients are negative and grow in magnitude (2014: -4.8 bp; 2015: -6.8 bp; 2019: -13.3 bp).

6.3 Robustness

I summarize the robustness exercises here; full results are in the Appendix.

Alternative estimators. The doubly robust inverse probability weighted (DRDID) estimator of Sant'Anna and Zhao (2020), residualized by lender and industry × quarter fixed effects with 999 lender block-bootstrap replications, yields an ATT of about 13 bp for the 7v12 binary sample, significant at the 5% level with a confidence interval bounded away from zero (untabulated), broadly consistent in sign and magnitude with the baseline.

Symmetric windows. Restricting to a symmetric three-year window around the reform (2011Q1–2016Q4) yields a reduction of about 10.8 bp, significant at the 1% level (Table IA.11). The stability across windows rules out a long-run secular trend masquerading as a treatment effect.

Lender × year fixed effects. Replacing additive lender fixed effects with lender × year interactions attenuates the 7v12 estimate to about 7.8 bp, roughly three-quarters of the 10.7 bp baseline, though the estimate is now imprecise (untabulated). Most SBA lenders operate in a single state, so lender × year effects approximate state × year effects and absorb the identifying variation. For loans between $350K and $1M, the coefficient is about 11.5 bp, closer to the baseline magnitude though still imprecise.

Sample and inference robustness. The continuous all-states spread specification remains significant excluding any single state other than Texas (Table IA.8; see Texas sensitivity paragraph below). Routine inference checks, including the wild cluster bootstrap of MacKinnon et al. (2023), are tabulated in the Internet Appendix (Table IA.12). HonestDiD sensitivity analysis (Rambachan and Roth, 2023) is uninformative for the spread outcome because the quarterly estimates are too imprecise to bound the post-treatment effect under calibrated violations of parallel trends; the parallel-trends case for spreads rests on the annual Wald test, which does not reject parallel pre-reform trends (Figure 4).

Excluding Ohio. Ohio raised its homestead exemption sharply in March 2013, roughly nine months before the reform, raising the concern that a state-specific exemption change near the reform window, rather than the reform itself, drives the continuous gradient. Dropping Ohio from the all-states continuous specification leaves the gradient essentially unchanged and, if anything, slightly stronger: the coefficient is about -1.60 bp per unit of log effective exemption, significant at the 5% level (Table 8), against the -1.47 bp baseline. The Ohio exemption jump does not drive the result.

Texas sensitivity. The design-robust spread estimate is the interior gradient, which excludes the unlimited-exemption states and does not lean on Texas; the binary and the full-distribution continuous specification, by contrast, do. Table IA.7 reports leave-one-state-out sensitivity. Dropping Texas renders the binary DiD insignificant, with the coefficient falling to about 1.4 bp, reflecting Texas's unique status as the only state where the constitutional prohibition on home-equity business liens makes the lender's exclusive dependence on business collateral exact (Zevelev, 2021). A Texas-only binary DiD (Texas versus the twelve low-exemption states; untabulated) yields the largest estimates, an upper bound on the treatment effect: a spread reduction of about 18 bp under both the baseline specification and the NAICS2 $\times$ quarter specification with home price controls, both significant at the 1% level, and the pre-trend Wald test does not reject parallel pre-reform trajectories. The continuous specification, by contrast, loses significance when Texas is excluded (Table IA.8; quantified below). Excluding all seven unlimited-exemption states, the continuous gradient remains significant at the 1% level across the 42 finite-exemption jurisdictions (83,841 loans; Figure 6), showing that the design-robust effect is driven by the interior of the exemption distribution, not by the Texas-versus-zero-HE contrast. A full continuous leave-one-state-out analysis (Appendix Table IA.8) confirms that no single state drives the continuous gradient. The Texas exclusion attenuates the continuous coefficient and renders it insignificant (Table IA.8), reflecting Texas's status as the dominant unlimited-exemption mass in the continuous gradient. The remaining 48 exclusions all produce negative coefficients significant at the 10% level or stronger, with the Florida exclusion most negative and the Pennsylvania exclusion least negative among the significant set (Table IA.8). The continuous estimate, weighted across the full exemption distribution, is an average marginal effect that includes the attenuation in unlimited-exemption states permitting voluntary pledging.

Within-borrower test. Adding borrower fixed effects to the spread DiD yields a within-borrower estimate of about $+2.5$ bp for the 7v12 binary (Table IA.14, column 2), small, opposite in sign to the cross-sectional gradient, and statistically indistinguishable from zero. The estimate is also imprecise: its confidence interval is wide enough to admit repricing of roughly twenty basis points in either direction, so the test cannot confirm a precise within-borrower zero, and I do not read it as affirmative evidence of within-borrower invariance. The reading instead rests on the design: the cross-sectional gradient is identified across borrowers, and this underpowered within-borrower test shows no offsetting movement. For a borrower who stays with the same lender across the reform, the spread shows no detectable change, consistent with that borrower's price being disciplined by what a competitor would offer, where the competitor prices off pooled borrower-type information the reform does not touch. Read together, the cross-sectional gradient and the absence of detectable within-borrower repricing place the reform's pricing effect at the point of new-match formation rather than in the repricing of existing relationships, though the borrower-fixed-effects design cannot separate this reading from a generic pricing stickiness in which incumbents simply do not reprice existing relationships.

Falsification. Table 9 examines whether the reform changed borrower selection or program composition rather than collateral valuation. The repeat-borrower share at the state-quarter level is unchanged, the within-borrower probability of switching from Standard to Express is unchanged, and the fixed-rate share at origination is unchanged (Table 9). The Express-program spread placebo, restricted to Express loans above $150K to avoid the October 2013 Express fee waiver, is statistically indistinguishable from zero (Table 9); Express loans face minimal collateral requirements and should not respond to the SOP collateral-valuation reform. Charge-off rates show no differential change at the reform boundary (Appendix Table IA.10), supporting the maintained assumption that default probability is exogenous to the collateral valuation methodology.

The pricing results establish that spreads moved with the homestead exemption, the design's source of variation. The loan-level channel behind that gradient is less sharply identified: the effect is flat across loan sizes, and within the matched sample the movable-collateral loans where business assets dominate recovery show no net spread reduction. The reduction surfaces among the immovable-collateral loans the UCC match does not cover. I now turn to the lender adaptation that accompanies this gradient: how lenders reshaped the collateral they take in response to the standardized rule.

7 Results: Lender Adaptation

7.1 Collateral Composition Shift

Where lenders cannot fall back on home equity, the adaptation logic predicts that they rebuild the asset-class information the reform took away by filing more detailed, asset-specific collateral descriptions, while where home equity provides a recovery buffer, lenders instead file generic blanket liens. The reasoning is that standardizing valuation strips the lender's private read on business-asset recovery value: a lender whose recovery depends on those assets rebuilds it through precise itemized filings, while a lender with a home-equity cushion takes the cheapest route to perfecting its claim, a blanket lien. I test this using UCC filing data matched to SBA loans across California, Florida, and Colorado.

Table 10 reports summary statistics for the UCC-matched sample. Panel A shows reasonable loan-level balance (all normalized differences below 0.25 except the corporation indicator (0.40) and the fixed-rate indicator (0.25)). Panel B reveals large pre-reform level differences in collateral filing practices (blanket scope ND $= 1.03$, filing intensity index ND $= -0.82$), reflecting the distinct legal environments that motivate the research design. The event studies in Appendix Figures IA.2–IA.3 and Figure 5 show that pre-reform *trends* were parallel despite these level gaps. Appendix Table IA.9 reports the matched-sample composition by state and reform period: within-regime normalized differences for spread, loan size, and term are all no greater than 0.36 in absolute value, indicating that the matched-sample composition is stable pre-to-post within each state group and that matched-sample inference is not driven by selection on observables.

Table 11 reports the collateral composition results from estimating equation (4) on the UCC-matched subsample (18,341 classified loans; approximately 530 unique lender clusters). The sample comprises Florida loans (treated) and California and Colorado loans (control). Because this design compares one unlimited-exemption state (FL) against two limited-exemption states, I present the continuous treatment specification as the primary result and the binary comparison as a robustness check.[23]

Collateral composition (extensive margin). Table 11 reports the extensive-margin response on the full UCC-classified sample ($N = 18{,}341$). Two DiD coefficients characterize the shift. First, the blanket lien indicator rises by 5.4 pp in Florida relative to CA+CO, significant at the 5% level (col 3): Florida lenders used blanket-form UCC statements more frequently post-reform. Second, the scope score (a 0–5 count of floating-lien clauses such as after-acquired property, proceeds, and all-locations language) rises by about 0.28 in the binary comparison (col 7) and by about 0.058 per unit of log-exemption in the continuous specification (col 8), both significant at the 1% level. The aggregate scope score increase is driven mechanically by the extensive-margin shift toward blanket-form filings: blanket liens have higher scope scores than itemized filings by construction, so Florida's higher blanket-form rate translates directly into a higher aggregate scope score. These extensive-margin results are the firmest-signed piece of the adaptation prediction: where

[23]The control group here (CA and CO at $\sim\$75$K) differs from Design 1's low-exemption set; the continuous specification bridges the two designs (Section 5.3).

the homestead exemption shields the borrower's home and other personal assets from a deficiency claim, lenders broaden their business-asset claim to maintain collateral coverage. The remaining columns (N categories, equipment) are directionally consistent with broadening but statistically insignificant; the Fixtures/RE-related coefficient is negative and insignificant, consistent with substitution away from real-estate collateral.

The within-R^2 values (0.004–0.014) reflect substantial noise in individual filing-level data; the significant coefficients are identified from the large sample (18,341 loans) rather than strong model fit. Under additive lender, quarter, two-digit NAICS industry, and county fixed effects, matching Equation (4), the coefficients are stable across the controls included, indicating that the collateral adaptation pattern reflects the reform rather than lender-level drift.

Filing intensity (intensive margin). Table 12 reports the intensive-margin response on the full UCC-classified sample ($N = 18{,}341$), avoiding the bad-control problem of conditioning on the post-treatment blanket-lien indicator. Three DiD coefficients characterize how the form of post-reform filing broadening differs across the homestead-exemption regime. First, the blanket scope indicator (one for floating-lien filings that combine all-assets, after-acquired property, proceeds, and at-all-locations language, so the lien attaches to a freely replaceable asset pool rather than to identified pledged assets; zero otherwise) falls by about 0.130 in Florida relative to CA+CO in the binary comparison and by about 0.035 per log-exemption unit in the continuous specification, both significant at the 5% level (Table 12, cols 1–2): post-reform, CA+CO lenders pushed a larger share of their filings toward generic floating-lien boilerplate, while Florida did not. Second, the specificity score, which counts asset-specific descriptors (serial numbers, make/model, condition language) at the filing level, rises by about 0.176 in Florida relative to CA+CO in the binary comparison and by about 0.036 per log-exemption unit in the continuous specification, significant at the 1% and 5% levels respectively (Table 12, cols 3–4). Third, the composite filing intensity index, redefined as filing-level information density across the full UCC sample, rises by about 0.300 standard deviations in Florida in the binary comparison and by about 0.074 per log-exemption unit in the continuous specification, both significant at the 1% level (Table 12, cols 5–6).[24,25]

[24]The composite filing intensity index (constructed in Section 4.2) is here computed on the full UCC sample, measuring filing-level information density rather than within-blanket intensity. Of its components, the asset-specificity score is the most directly interpretable and the firmest piece of the adaptation prediction; the composite index summarizes the three and moves in the same direction. Because the three outcomes are correlated components of the same underlying construct, no family-wise error rate adjustment is applied.

[25]Diagnostic on filing-mix composition: the non-blanket filing share rose 7.2 pp in CA+CO and 1.6 pp in Florida post-reform, a DiD of -5.6 pp. The filing-intensity-index coefficient absorbs this differential without

The intuition is as follows. Treatment shifts the entire UCC filing distribution rather than only the form of blanket-lien content. Florida, the high-exemption state where the homestead exemption shelters the borrower's home from a deficiency claim, responds by expanding both blanket adoption (extensive margin, about 5.4 pp; preceding paragraph) and asset-specific descriptors. CA+CO, the low-exemption states where home equity provides a recovery buffer, respond by adopting generic floating-lien boilerplate within the filings they do produce and by adding more itemized non-blanket filings. A floating lien attaches to a replaceable pool of business assets the borrower can dispose of in the ordinary course, whereas an asset-specific pledge fixes on identified assets the borrower cannot sell without the lender's consent. Both groups broaden post-reform; the form of broadening differs by exemption regime. The extensive-margin piece is the firmest-signed part of the adaptation prediction: where the home provides no recovery buffer, the lender's return to broadening its business-asset claim is highest, and the post-reform rate of blanket-lien adoption rises in the homestead exemption. The intensive-margin piece, the specific form the broadening takes, carries a revealed-preference hedge; additive lender fixed effects absorb reform-invariant lender heterogeneity and isolate the within-lender post-reform shift on specificity and blanket scope.

Read through the selection-and-enforcement framework of Section 3, these patterns map onto the two defenses. The specificity gradient is the clean test of the enforcement margin: filings grow more asset-specific where the exemption is higher, so enforcement on the movable book rises in the same states where tilting into real estate grows dear. Selection weakens with the exemption in either regime, so the always-signed prediction is the one the framework leans on. The rise in specificity is consistent with the diversification regime, in which the borrowers who can pledge real estate hold movable collateral that is less, not more, enforceable than the natural prior would suggest; but because this reading is inferred from the specificity gradient itself and runs against that prior, I treat the specificity rise as a descriptive fact the diversification regime can rationalize rather than as evidence for it. The composition evidence is consistent with selection weakening where the exemption is high: the broadening toward movable, blanket-form filings reported above is the empirical face of the book moving off real estate, and the negative real-estate-related coefficient points the same way, though it is imprecisely estimated. Because the diversification reading is inferred from the specificity gradient rather than from independent evidence on the borrower correlation, and the filing data cover a single high-

sign change because all three component outcomes (blanket scope indicator, specificity score, N categories) move in the directions reported above on the full UCC sample, confirming that the construct is robust to the small filing-mix shift.

exemption state, I treat it as an ex-post interpretation the evidence supports, not a separately identified parameter or a test of the regime.

The filing-intensity-index event study (Figure 5) shows pre-2013 coefficients close to zero and a sharp break at the reform date. The joint Wald test comfortably fails to reject parallel pre-reform trajectories on all three outcomes (blanket scope, specificity, and the filing intensity index), as shown in the event-study figures (Figure 5 and Appendix Figures IA.2–IA.3).

Robustness to lender continuity. A residual concern is that the headline pattern reflects compositional entry of new UCC filers post-reform rather than incumbent behavioral change. Restricting to the 141 lenders who filed at least one UCC statement in both the pre- and post-reform periods ($N = 17{,}716$) leaves all three intensive-margin coefficients essentially unchanged, each within roughly 7% of the headline estimate and significant at the 5% level or stronger (Table IA.3). The findings reflect incumbent behavioral change among lenders with continuous UCC activity, not differential entry into the UCC-filer pool.

Filing behavior (extensive margin). The pooled CA+CO UCC match rate rose from 45% to 59% post-reform (14 percentage points), while Florida's remained stable at 84–86%. The DiD is about 13.8 percentage points and significant at the 1% level (Table 11, cols 1–2, estimated on the full >\$350K CA/CO/FL sample including unmatched loans, $N = 29{,}618$; the composition outcomes in cols 3–12 condition on the UCC-classified subsample, $N = 18{,}341$): limited-exemption lenders expanded UCC filing significantly more than unlimited-exemption lenders. The pattern is consistent with limited-HE lenders under-filing pre-reform, when home equity provided a recovery buffer that reduced the return to business-asset perfection, and then filing broadly once the reform eliminated the information cost of claiming collateral credit. Conditional on filing, unlimited-HE lenders broaden the legal claim perimeter through blanket-form filings paired with asset-specific descriptions, whereas limited-HE lenders broaden through generic floating-lien boilerplate language: the form of post-reform broadening differs by exemption regime, even though both groups expand their claim perimeter.

7.2 Filing Intensity and Spread Changes

Having established the collateral-specificity result, I now decompose it against pricing. If lenders who rebuild asset-class information post-reform pay for it through the filings they make, then within the UCC-matched sample the lenders who increase filing intensity should show smaller spread reductions than those who do not, because the filing cost partly offsets the rent the reform recovers. This is an interpretive decomposition of the adaptation result, not a separate test of it; the estimates are too imprecise to confirm pass-through. I examine it using triple interactions within the multi-state lender subsample (lenders operating in at least two of FL, CA, and CO), restricting to loans in the $350K–$1M tier.[26]

Table 13 reports three specifications. First, the loan-level blanket lien indicator interacted with the FL × Post DiD yields a directionally consistent but imprecise triple interaction of about -13.9 bp (Table 13): among Florida loans, those with blanket lien filings see a smaller spread increase than non-blanket loans. Second, the pre-to-post change in lender-level filing intensity produces a larger but imprecise coefficient. Lenders who increased filing intensity post-reform are associated with a roughly 19.9 bp smaller spread increase than lenders who did not, but the estimate is imprecise (Table 13). The base FL × Post coefficient for lenders who did not increase filing intensity is a spread increase of about 19.8 bp, significant at the 5% level, so FL lenders who increased filing intensity experienced a net spread change of essentially zero (Table 13). Among Florida loans, those whose lenders increased filing intensity saw essentially no net spread change while others did not; the difference is directionally consistent with lenders absorbing the recovered rent into filing cost, though the estimate is too imprecise to confirm. The wide standard error means the data are equally consistent with a large offset and with no offset at all.[27] The filing-intensity-change test isolates the reform-boundary margin by comparing the same lender's behavior before and after January 2014, rather than comparing structurally different lender types cross-sectionally.

These results connect the collateral composition evidence (Section 7.1) to pricing outcomes. The cross-sectional null from the previous framework, that the Treated × Post × HighEffort triple interaction is zero in the full blanket lien sample (Table 14), is not contradicted. The cross-sectional test compares high- versus low-intensity loans *within treatment groups*, where filing intensity levels reflect both pre-reform lender type and post-reform

[26] The $1 million upper bound narrows the sample to the dense mass of SBA Standard Guaranty loans and avoids noise from the small number of very large loans ($1M–$5M) whose pricing differs from the typical 7(a) loan. Table 11 (collateral composition) uses the full >$350K sample without the upper bound. All sample restrictions and variable definitions (filing intensity index, winsorization, lender activity filter) are inherited from the canonical `full_state_sample.rds` used in the main pricing specifications.

[27] The minimum detectable effect at 80% power and 5% two-sided significance is approximately 36 bp; observed power for the estimated effect of about -19.9 bp is approximately 34%. The test is therefore underpowered against an effect of the magnitude implied by the adaptation result; the imprecision should be read as a sample-size constraint of the three-state UCC subsample (8,389 loans), not as evidence of zero pass-through.

adjustment. The dynamic test here isolates the *change* in filing intensity at the reform boundary, which is the margin relevant for information-acquisition attenuation. One reading consistent with the data is that lenders who maintained or increased filing intensity post-reform absorbed the uncertainty-reduction benefit into information-acquisition costs, while those who reduced filing intensity (predominantly in CA and CO) passed it through as lower spreads. The triple interaction is too imprecise to distinguish this interpretation from a null (Table 13).

7.3 Florida versus California and Colorado

A natural question is why the FL versus CA+CO spread DiD is positive, with a small and insignificant point estimate of about 6 bp (untabulated), when the homestead-exemption logic would lead one to expect a negative effect. The raw spread changes show where this comes from: unconditional pre-post mean spreads (all loans, not regression-adjusted) fell by 49 bp in California and 43 bp in Colorado, but only 8 bp in Florida. I read this as a feature of the three-state UCC sample rather than a failure of the reform in Florida, and the collateral-specificity evidence offers a reading that is consistent with the pattern.

The UCC-matched sample consists entirely of movable-collateral loans, where the preset haircuts (50–75%) are less generous than the real estate haircut (85%) and where pre-reform valuation uncertainty was higher. In this setting, the net spread change depends on the balance between uncertainty reduction and information-acquisition cost. California and Colorado lenders, who benefit from a home-equity recovery buffer, reduced information acquisition post-reform (pooled filing intensity index: $+0.44 \rightarrow +0.08$; Table 10), accepting preset valuations and passing through the uncertainty reduction as lower spreads. Florida lenders, who rely primarily on business collateral for recovery, maintained asset-specific information acquisition (filing intensity index: $-0.35 \rightarrow -0.38$), preserving their private valuation advantage through precise asset-level filings. The information acquisition cost partially offset the uncertainty-reduction benefit, producing a near-zero net spread change.

The FL vs CA+CO raw spread comparison is therefore *consistent with* the collateral-specificity channel rather than evidence against the spread result; I treat it as descriptive context, not as an independent test of pass-through.

7.4 Collateral Requirement Shift

The reform's extensive margin provides a final piece of evidence on the credit channel. If preset haircuts change what loans require collateral, this affects the composition of the borrower pool reaching the collateral adequacy review.

The fraction of loans with collateral pledged declines by about 4.9 percentage points more in unlimited-HE states post-reform in the 7v12 binary specification, and by about 0.72 pp per unit of log-exemption in the continuous specification, both significant at the 5% level (Table 15). Post-reform, some loans in high-HE states no longer trigger the collateral adequacy requirement under the preset haircut schedule, either because the standardized valuation pushes the loan below the fully-secured threshold or because the preset rules reduce the information acquisition burden, allowing marginal cases to proceed without formal collateral pledges. This extensive-margin relaxation is not in tension with the intensive-margin rise in specificity, on the reading that the marginal loans freed from the requirement are those the preset schedule now deems adequately secured, whereas the rise in filing specificity occurs on the inframarginal loans where the lender does take and perfect business collateral; I offer this as an interpretation rather than a directly shown decomposition. Loan counts show no differential change in the 7v12 binary (Table 15), indicating that the reform altered collateral requirements rather than loan approval rates. The reform changed the terms of credit, not access to it.

Taken together, the results trace a reform whose first-order effect is an adaptation in how lenders take collateral, with the change in spreads following as a consequence. Lenders in unlimited-HE states, who cannot fall back on home equity, rebuilt the asset-class read through more asset-specific filings, while limited-HE lenders shifted toward generic blanket liens. This adaptation shows up in prices: spreads fell with the homestead exemption, the reduction surfacing among immovable-collateral loans and running to near zero among the movable-collateral loans where lenders invested most in information acquisition. The exemption gradient is identified, but the business-collateral-share channel behind it is not separately corroborated at the loan level, so it is one consistent reading rather than an established mechanism. The pricing and adaptation results are co-identified by the homestead exemption rather than linked by a single estimated loan-level channel; the adaptation-first ordering is the framework's reading, not a demonstrated causal chain. Within-borrower evidence is consistent with this picture: the spread for existing relationships shows no detectable change, in line with an outside option priced off borrower-type information the reform does not touch, so the cross-sectional gradient, identified across borrowers, places the reform's pricing effect in the formation of new matches rather than in within-borrower repricing. The collateral requirement shift indicates that the reform altered the terms of credit. I turn next to policy implications and the limits of the identification.

8 Discussion

8.1 Policy Implications

Collateral valuation standardization achieves its stated goals: the standardized haircut schedule (Section 2.3) replaced subjective, heterogeneous assessments with known, verifiable rules, reducing valuation uncertainty and improving consistency. But its first-order effect is to change how lenders take collateral, not only what they charge. Facing a rule that fixes the recognized value of business assets, lenders whose recovery cannot lean on home equity rebuild the lost asset-class read by filing more detailed, asset-specific liens, while lenders with a home-equity cushion take the cheapest route to perfection. The pricing change follows from this adaptation.

These benefits carry an offsetting cost that the reform's design cannot avoid. Lenders in unlimited-exemption states, where friction in accessing home equity limits the recovery buffer, face higher information acquisition costs as they invest in detailed collateral descriptions to reduce residual uncertainty on movable assets. This cost may be passed through to borrowers as higher spreads, partially offsetting the benefit of standardization; the information-cost pass-through triple interaction is directionally consistent with this mechanism but too imprecise to confirm it (Section 7.2), so the offset is a possibility the data are consistent with rather than an established cost, and the policy reading is correspondingly conditional. The relationship margin, by contrast, is not a cost of the reform at all: because the standardized rule reaches only business-asset valuation, the borrower-specific knowledge accumulated through servicing is untouched, so within-borrower prices show no detectable repricing and the pricing response surfaces, by design, in the cross-section of new matches.

These two effects have different policy implications. Permitting lender discretion (e.g., flexible valuation rules or supplementary appraisals) can partially offset the pricing cost by reducing the need for costly information acquisition. The allocative consequences are harder to reverse: because the relationship margin is left intact, the reform's pricing effect falls entirely on newly formed matches, and the value of relationship-specific knowledge inside ongoing relationships is unchanged whenever rule-based valuation displaces lender-specific judgment of business-asset value.

8.2 Alternative Interpretations

What the spread gradient identifies is the reduction in valuation uncertainty on business assets that the standardized rule delivers, not any single component of it. Before the reform, the lender's discretionary appraisal of business collateral carried both a private valuation advantage and the compliance risk that an unverifiable, subjective valuation would later be challenged; the preset, uniform schedule retires both at once. The two move together, each scaling with the fraction of expected recovery that comes from business collateral and each larger where the homestead exemption shields more of the borrower's other personal assets, so the spread DiD cannot separate the compressed discretion rent from the retired compliance premium. I therefore read the gradient as the price consequence of reduced valuation uncertainty, a bundle of the two, rather than as either alone.

Two structurally distinct mechanisms remain potentially observationally equivalent on the cross-sectional exemption-graded gradient, and separating them would require admission-margin or lender-entry data outside the present setting. The first is selection on default risk at the admission margin. If pre-reform rationing operated on collateral coverage rather than on screening intensity, the reform raises the recognized value of borrowers who pledge a large share of business collateral disproportionately, shifting the admitted pool toward lower default risk precisely where recovery concentrates on business assets. The falsification evidence is consistent with this concern being small in the present sample (charge-off rates and the repeat-borrower share are unchanged), but it is not ruled out. The second is supply-side competition with cross-lender heterogeneity in compliance cost. Where lenders differed most in how costly the old, subjective valuation was to defend, the reform compresses markups most; if that dispersion correlates positively with the homestead exemption, the spread gradient again rises with the exemption. Lender entry and exit show no differential response in the falsification table, but lender-level conduct heterogeneity unrelated to entry would not be detected by that test. Reduced valuation uncertainty, the bundle the design identifies, remains the maintained reading, with these alternatives acknowledged rather than ruled out. The broader regulatory-standardization literature on rule-based versus discretionary supervision provides additional context for both alternatives (Behn et al., 2022; Plosser and Santos, 2018; Agarwal et al., 2017).

8.3 External Validity

The SBA 7(a) program carries a 75% federal guarantee, so the lender bears only 25% of any net loss after collateral liquidation. A competitive break-even spread scales with this residual loss share, so the lender's exposure to collateral recovery, and therefore to how that collateral is valued, is muted by the guarantee. In unguaranteed commercial lending, where the lender bears the full residual risk, the pricing effects could be substantially larger; scaling by the residual loss share gives a mechanical, arithmetic upper bound on the order of four times the estimate under a single linear assumption, not a structural prediction the data identify. The contribution is therefore the existence and direction of the valuation-method channel, which the design establishes cleanly, not a magnitude that ports to unguaranteed lending. The guarantee is, however, precisely what makes the SBA setting informative: because the SBA sets the collateral adequacy rules, the reform provides a clean policy experiment in which the treatment is the valuation methodology, not the underlying asset values. SBA 7(a) lenders are predominantly community banks and CDFIs, whose relationship intensity exceeds that of large commercial banks on average; the scope for valuation discretion is therefore narrower where lending is already standardized. The mechanism should nonetheless generalize to other standardization episodes in financial regulation, including Basel risk-weight standardization and automated credit scoring, where similar tradeoffs between consistency and lender-specific information arise.

8.4 Limitations

The analysis conditions on approved loans. Borrowers who were denied credit under the new rules are unobserved, and compositional shifts in the approval pool may contribute to the estimated effects. The SBA guarantee structure limits the scope for extensive-margin selection (loan counts show no differential change), but the guarantee does not eliminate it.

The binary specification is sensitive to Texas, the only state where the constitutional prohibition on home equity business liens makes the treatment exact. The continuous specification, which exploits the full exemption distribution and does not depend on any binary classification, provides the more robust evidence.

The UCC evidence covers three states (California, Colorado, Florida). The movable/immovable proxy relies on UCC match status rather than direct observation of pledged asset types. Within the three-state sample, the proxy is supported by summary statistics: matched loans carry higher spreads, shorter maturities, and lower fixed-rate shares. However, the proxy cannot be validated outside these states.

The filing-intensity-change moderator is endogenous to the reform. Lenders choose how much to invest in information acquisition in response to the new rules, and that choice may correlate with unobserved lender characteristics that also affect pricing. The information-cost pass-through interaction should be interpreted as a descriptive decomposition of the treatment effect, not a causal mediation estimate.

A related limitation concerns the gap between the filing intensity index and the private information it stands in for. The valuation uncertainty the reform reduces rides on the lender's pre-reform discretionary read of recovery value on specific business assets. The filing intensity index, by contrast, captures observable properties of the filing: how many Article 9 categories are listed and whether the filing uses generic or itemized language. The two coincide when lenders who know more about their collateral produce more granular filings, but the correspondence cannot be tested within filing data alone. To the extent that private knowledge and filing granularity diverge (for instance, if lenders with deep asset knowledge file generically because the filing cost is trivially low regardless of what they know), the index understates the lender's actual private information and the information-cost pass-through interaction understates the rent-compression component of the reduced valuation uncertainty. The imprecise pass-through pattern is therefore consistent with the index tracking private knowledge but does not establish it.

9 Conclusion

This paper studies how lenders adapt the collateral they take, and how pricing and credit allocation follow, when a standardized, preset collateral valuation schedule replaces lender discretion. The reform reached two distinct margins of lender private information, and the two met opposite fates. The first is asset-class information: the lender's discretionary read on how much business collateral would recover in default. Standardizing valuation strips this advantage directly, but the lender can rebuild it through observable, costly, asset-specific filings, so its pricing footprint is recoverable. The second is relationship information: what the incumbent knows about the borrower's type. The reform standardizes only asset valuation and leaves this margin untouched, so it survives intact and leaves within-borrower prices without detectable movement; the reform's pricing effect surfaces instead, by design, in the cross-section of newly formed borrower-lender matches rather than in the repricing of ongoing relationships.

The January 2014 SBA reform generated three sets of findings that trace these two margins. First, spreads fell where homestead exemptions are higher and business collateral is the lender's primary recovery channel; the more a state's homestead exemption shields the borrower's home and other personal assets from a deficiency claim, the more recovery leans on business assets, the more pricing weight the lender's pre-reform discretionary appraisal carried, and the larger the reduction once the preset, uniform schedule removed that valuation advantage and the compliance risk that rode on it. The reduction is driven by immovable (real estate) collateral and attenuated among movable-collateral loans, where lenders rebuilt the lost asset-class information through more detailed, asset-specific filings. Second, where lenders cannot fall back on home equity, they file more detailed, asset-specific collateral descriptions to rebuild that information; where home equity provides a recovery buffer, they instead take the cheapest route to perfection and file generic blanket liens. The extensive-margin reallocation toward itemized filings is firm; the intensive-margin form-of-broadening is read through revealed preference and the triple interaction on the cost pass-through is imprecise. Third, the pricing gradient is, by design, a cross-sectional new-borrower phenomenon rather than within-borrower repricing: the gradient is identified across borrowers, and the within-borrower estimate, small and too imprecise to affirm invariance on its own, shows no detectable change for a borrower who stays with the same lender across the reform, consistent with that price being disciplined by what a competitor would offer, where the competitor prices off pooled borrower-type information the reform does not touch. Set against the negative cross-sectional gradient, this locates the reform's pricing effect at the formation of new borrower-lender matches rather than in the repricing of ongoing relationships.

These results show that valuation standardization affects not only pricing but the full ecosystem of lender and borrower responses, and that its reach depends on which private-information margin it touches. Asset-class private information about the quality of business collateral is partially recoverable through observable, asset-specific filings and shows up as an attenuated pricing effect for movable-collateral loans. Relationship-specific private information about borrower type is left untouched by a reform that standardizes only asset valuation, so existing relationships show no detectable within-borrower repricing and the pricing gradient appears, by design, among newly formed matches. Standardization's costs thus depend on the capacity of regulated institutions to rebuild the recoverable asset-class margin, and its allocative footprint falls on the formation of new matches, where the relationship margin the reform never reached still shapes who newly lends to whom.

Section 8 discusses limitations, including selection into approved loans, Texas sensitivity in the binary design, the three-state scope of the UCC evidence, and the endogeneity of the filing-intensity moderator. Matching SBA loans to firm-level outcomes via Census data would reveal whether the new-match pricing effect translates into real effects on business activity. Other standardization episodes in financial regulation (Basel risk weights, automated underwriting, algorithmic credit scoring) may generate similar tradeoffs between consistency and information; cross-market comparisons would test the generality of the dual-channel mechanism shown here.

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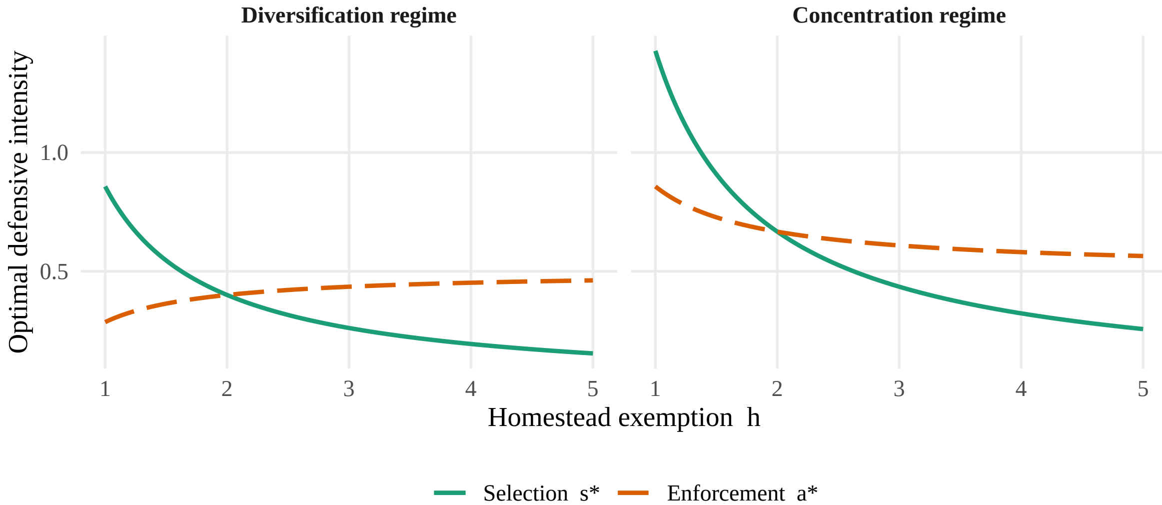


Figure 1: Optimal selection s^* and enforcement a^* across the homestead exemption h, under the two regimes of the framework in Section 3. In the diversification regime (left, $\gamma_0 < 0$), where the lender's two recovery sources are negatively correlated, a higher exemption lowers selection, the real-estate tilt, while it raises enforcement, lien specificity, so the lender substitutes one defense for the other; in the concentration regime (right, $\gamma_0 > 0$), where the sources are correlated, both weaken together. Selection always falls with the exemption (Proposition 1); the enforcement gradient carries the opposite sign of the borrower correlation γ_0 (Proposition 2). The curves are a schematic of the model's comparative statics under illustrative parameters, not calibrated to data.

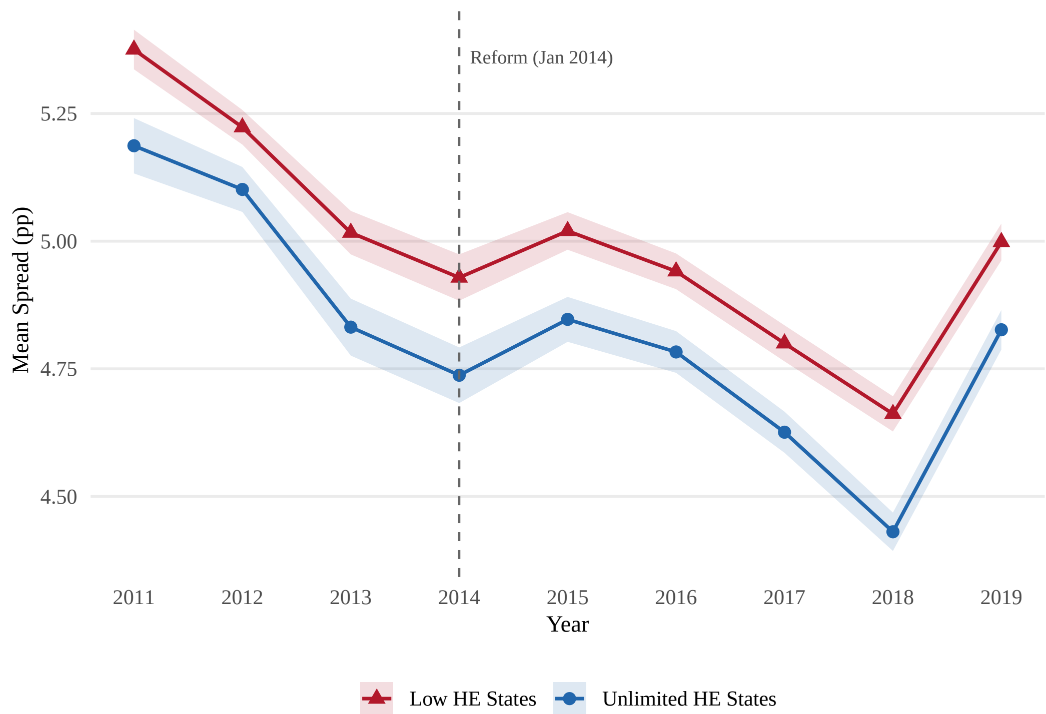


Figure 2: Mean Spread by Treatment Group and Year. Shaded bands show 95% confidence intervals around annual group means.

Table 2: SBA Collateral Haircut Schedule (Post-January 2014)

Asset Type	Valuation Rule	Lender Discretion
New machinery & equipment[a]	75% of purchase price, minus prior liens	None (mechanical)
Used machinery & equipment[a]	50% of NBV (80% with OLV appraisal), minus prior liens	Limited (OLV option)
Real estate[b]	85% of market value	Preserved (appraisal)

Notes: NBV = net book value (original price minus depreciation). OLV = orderly liquidation value. Source: SBA SOP 50 10 5(F), effective January 1, 2014. The business-asset haircuts remained unchanged through SOP 50 10 5(K), April 2019, covering the entire sample period. For real estate, the 85% haircut is applied to a market-value appraisal that is functionally equivalent to the pre-reform OLV; lender discretion over real estate valuation is therefore preserved. [a] The SOP refers to "machinery and equipment"; furniture and fixtures were implicitly included under this category during the sample period and were carved out as a separate asset class in SOP 50 10 6 (October 2020, post-sample). [b] The SOP refers to "real estate" without distinguishing improved from unimproved; the distinction was formalized in SOP 50 10 6 (October 2020, post-sample). Unimproved real estate was addressed in later SOPs and is excluded from this table.

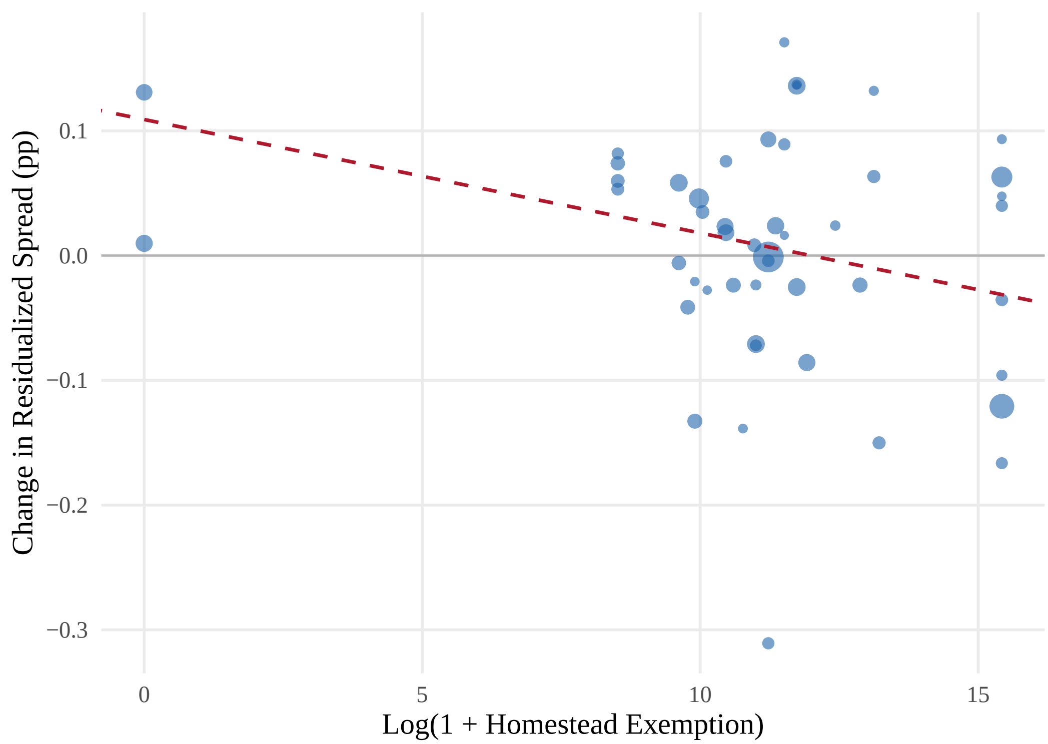


Figure 3: Pre-to-post reform change in residualized spread by state, plotted against log homestead exemption. Spreads are residualized on loan-level controls, lender, NAICS2 × quarter, and county fixed effects. Each point is a state; size reflects sample size. The dashed line shows the slope from the continuous DiD (Table 6, column 1).

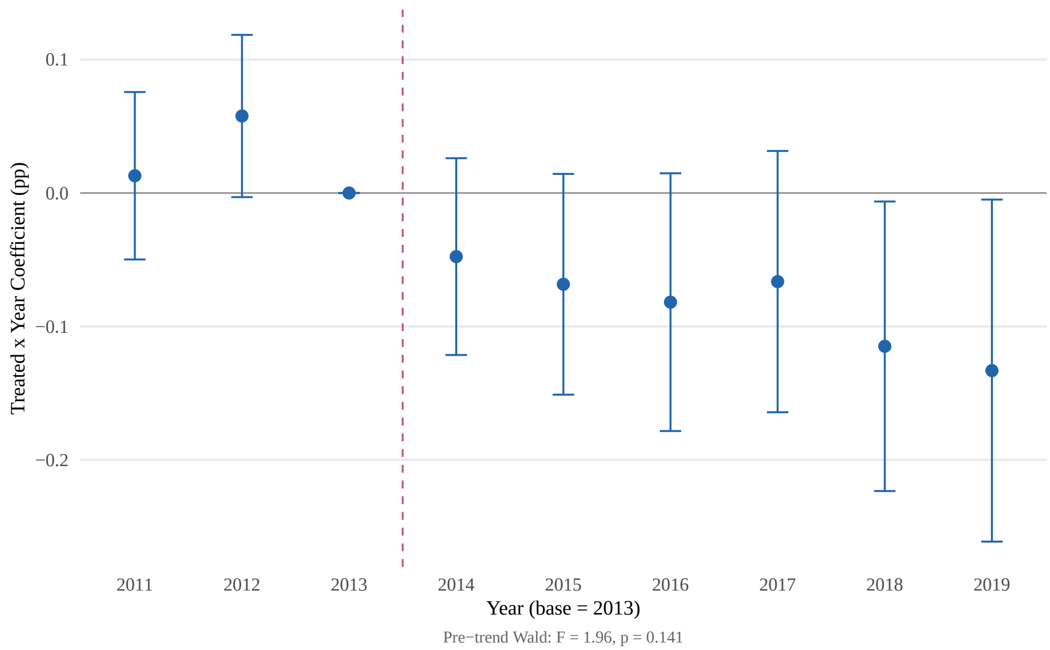


Figure 4: Spread Event Study: Treated × Year Coefficients. Year-by-year coefficients on Treated × Year interactions from the binary difference-in-differences specification, with the loan spread in percentage points as the dependent variable. Treated states are the seven unlimited-exemption states; control states are the twelve low-exemption states. The reference year is 2013, so each point is the spread gap relative to 2013; bars show 95% confidence intervals clustered at the lender level. Negative post-reform coefficients indicate spreads falling in treated relative to control states. The joint pre-trend Wald test is reported in Table IA.6. Sample: SBA 7(a) Standard Guaranty loans, 2011Q1–2019Q4.

Table 3: Summary Statistics

	High HE (above median)		Low HE (below median)		Norm.
	Pre	Post	Pre	Post	Diff.
Panel A: All loans (>$350K Standard Guaranty, all states)					
Loan amount (in $1,000)	1112.65	1229.36	1108.75	1232.64	0.005
	(882.74)	(995.24)	(868.28)	(993.61)	
Term (months)	214.21	214.20	206.33	207.40	0.079
	(91.35)	(90.81)	(90.68)	(89.33)	
Spread (pp)	4.93	4.55	5.12	4.81	-0.226
	(1.11)	(1.21)	(1.00)	(1.09)	
Fixed rate	0.206	0.256	0.183	0.208	0.103
Relationship	0.088	0.079	0.085	0.078	0.005
Previous SBA	0.151	0.135	0.141	0.130	0.017
Collateral	0.925	0.923	0.880	0.924	0.045
Corporation	0.872	0.906	0.934	0.948	-0.171
Franchise	0.082	0.161	0.107	0.198	-0.080
Homestead exemption ($1,000)	1580.4	1651.0	23.2	22.6	1.012
	(2232.2)	(2257.7)	(18.1)	(16.8)	
Observations	15,748	48,761	11,518	27,852	
Panel B: Repeat borrowers ($350K–$1M, all states)					
First loan amount (in $1,000)	605.57	619.43	605.16	614.66	0.022
	(182.97)	(182.81)	(181.54)	(181.90)	
First loan term (months)	189.88	196.01	184.94	187.15	0.084
	(94.53)	(91.84)	(91.43)	(89.13)	
First loan spread (pp)	5.04	4.75	5.21	4.93	-0.188
	(1.03)	(1.19)	(0.90)	(1.09)	
Fixed rate	0.175	0.228	0.148	0.219	0.055
Relationship	0.165	0.196	0.219	0.210	-0.073
Previous SBA	0.251	0.260	0.304	0.285	-0.082
Collateral	0.906	0.906	0.822	0.909	0.111
Corporation	0.892	0.895	0.963	0.962	-0.268
Franchise	0.043	0.154	0.083	0.158	-0.031
Number of loans	2.22	2.10	2.12	2.10	0.074
	(0.63)	(0.36)	(0.42)	(0.32)	
Observations	702	1,283	540	685	

Notes: Panel A reports loan-level statistics for all >$350K Standard Guaranty loans across all states with non-missing homestead exemption data, 2011Q1–2019Q4. High HE = states with exemption at or above the sample median; Low HE = states below the median. Panel B reports first-loan characteristics for repeat borrowers ($350K–$1M). Normalized differences computed as $(\bar{x}_T - \bar{x}_C)/\sqrt{(s_T^2 + s_C^2)/2}$.

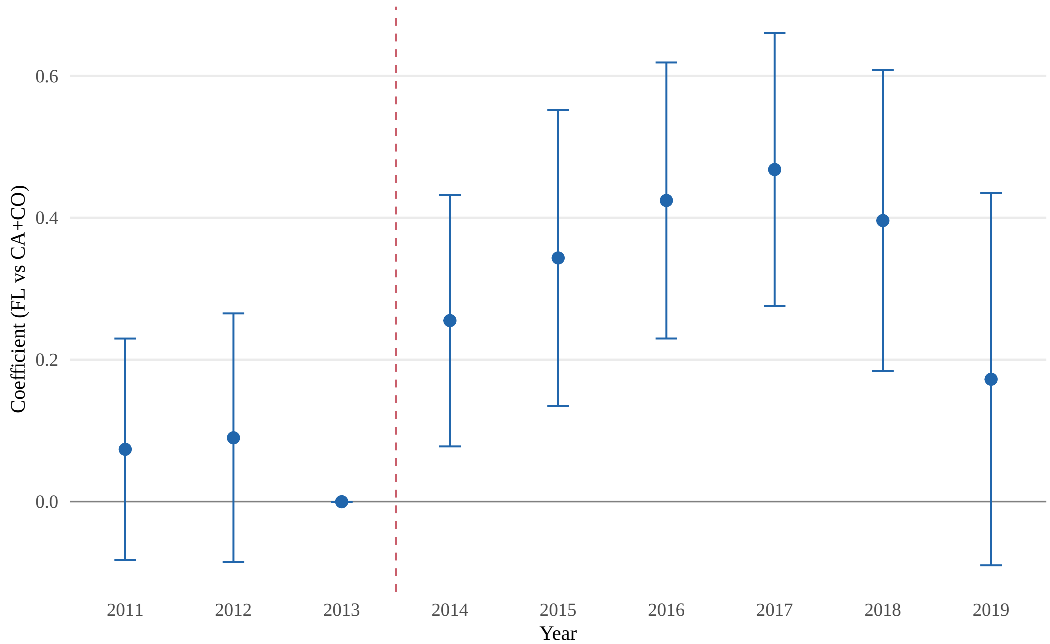


Figure 5: Event Study: Filing Intensity Index, Full UCC-Classified Sample. Reference year is 2013. Wald test for pre-trends: $F = 0.61$, $p = 0.542$.

Table 4: Balance Table: Pre-Reform Repeat Borrowers ($350K–$1M)

	Treated		Control		Norm.	
	Mean	SD	Mean	SD	Diff.	Balance
First loan amount ($)	606709.66	182997.09	598758.99	177317.49	0.044	
First loan term (months)	171.19	91.08	185.06	91.13	-0.152	
First loan spread (pp)	5.02	1.05	5.32	0.83	-0.319	†
Fixed rate	0.23	0.42	0.11	0.31	0.323	†
Relationship	0.14	0.34	0.21	0.41	-0.192	
Previous SBA	0.20	0.40	0.31	0.46	-0.263	†
Collateral	0.92	0.27	0.82	0.38	0.292	†
Corporation	0.93	0.26	0.94	0.23	-0.066	
Franchise	0.05	0.21	0.10	0.30	-0.201	
Number of loans	2.17	0.55	2.11	0.39	0.131	
Observations	176		278			

Notes: Pre-reform (entry before 2014Q1) repeat borrowers, $350K–$1M. Normalized differences computed as $(\bar{x}_T - \bar{x}_C)/\sqrt{(s_T^2 + s_C^2)/2}$. † indicates |Norm. Diff.| > 0.25.

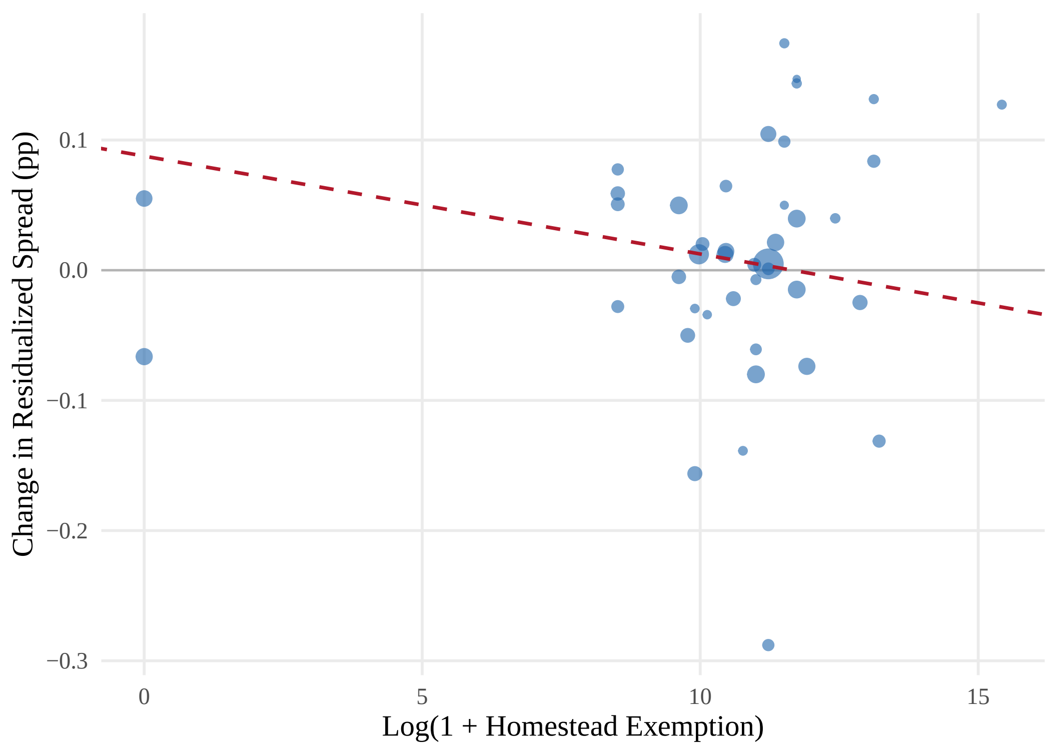


Figure 6: Pre-to-post reform change in residualized spread by state, excluding the seven unlimited-exemption states. Spreads are residualized on loan-level controls, lender, NAICS2 $\times$ quarter, and county fixed effects. The dashed line shows the continuous DiD slope ($\beta = -0.0075$, $p = 0.009$, 42 states, 83,841 loans). The gradient is significant at 1% from the interior of the exemption distribution alone.

Table 5: Variable Definitions: Empirical

Variable	Definition
Spread	Interest rate minus matched-maturity Treasury yield (percentage points). Variable-rate: rate − 3-month T-bill. Fixed-rate: rate − piecewise-linearly interpolated Treasury yield at the loan's maturity.
Treated	$=1$ for unlimited-HE states (AR, FL, IA, KS, OK, SD, TX); $=0$ for low-HE states (AL, GA, IL, IN, KY, MD, MO, NJ, PA, TN, VA, WY; exemption $\leq$ \$23K).
Post	$=1$ for loans approved on or after January 1, 2014.
$\log(1+\text{exemption})$	Log dollar homestead exemption; continuous treatment in the all-states specification.
Growth	$\log(L_2/L_1)$ for repeat borrowers with at least two SBA loans within the \$350K–\$1M tier.
PreviousSBA	$=1$ if borrower has any prior SBA loan from any lender; proxy for $\delta_r > 0$.
UCC Matched	$=1$ if loan matched to a UCC filing. Because UCC Article 9 covers only personal (movable) property, serves as a movable collateral proxy.
Blanket Lien	$=1$ if UCC filing pledges "all assets" of the debtor.
Scope Score	Weighted count of floating-lien clauses in the UCC filing (0–5 scale).
Effort Index	Standardized composite within blanket liens; combines inverse scope, specificity score, and number of distinct Article 9 categories.
Effort Change	Pre-to-post change in lender-level mean effort index within the blanket-lien subsample.
Collateral Pledged	$=1$ if the SBA record indicates collateral was pledged.

Notes: Variables constructed from SBA administrative loan records, UCC filing data, and the homestead exemption panel (Section 4). The filing intensity index and lender-level filing-intensity change variables are defined only for the UCC-matched blanket-lien subsample.

Table 6: Spread Difference-in-Differences: Continuous Homestead Exemption (All States)

	Panel A: Full Sample				Panel B: By Loan Size		
	(1) Baseline	(2) Additive FE	(3) + HPrice	(4) Rank	(5) \$350K–500K	(6) \$500K–1M	(7) \$1M–2M
HE × Post	-0.0147^{**}	-0.0151^{**}	-0.0141^{**}	-0.0927^{**}	-0.0196^{*}	-0.0161^{**}	-0.0103
	(0.0068)	(0.0069)	(0.0068)	(0.0384)	(0.0100)	(0.0074)	(0.0089)
Observations	103,476	103,506	102,301	103,476	21,985	38,075	26,222
Within R^2	0.4166	0.4178	0.4162	0.4167	0.4315	0.4185	0.3667
Lender FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
NAICS2 × Quarter FE	Yes	No	Yes	Yes	Yes	Yes	Yes
Quarter FE (additive)	No	Yes	No	No	No	No	No
County FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Home price control	No	No	Yes	No	No	No	No

Notes: Dependent variable is the loan spread (percentage points over matched-maturity Treasury). Sample: >\$350K Standard Guaranty loans across all states with non-missing homestead exemption, 2011Q1–2019Q4. Column (1) is the preferred specification with NAICS2 × quarter fixed effects absorbing industry-time shocks. Column (2) replaces NAICS2 × quarter with additive quarter FE as a robustness check. Columns (3)–(7) all use the preferred NAICS2 × quarter specification: column (3) adds a county-level home price control; column (4) replaces log(effective exemption) with a rank-based measure (0–1 scale); columns (5)–(7) restrict the sample by loan-amount tier. All columns include controls for log loan amount, term, fixed-rate indicator, relationship lending, previous SBA, collateral, corporation, revolver, and franchise indicators. Standard errors clustered at the lender level. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 7: Loan-Size Heterogeneity: Spread DiD by Loan Amount (7v12 Binary)

	(1) \$350K–\$1M	(2) >\$1M–\$2M	(3) >\$2M
Treated × Post	-0.1115^{***} (0.0426)	-0.0898^{*} (0.0468)	-0.1009^{*} (0.0537)
Observations	24,851	11,068	6,888
Within R^2	0.3911	0.3423	0.3479
Sample	7v12 Binary	7v12 Binary	7v12 Binary
Controls	Yes	Yes	Yes
FE: Quarter + Lender + Industry×Quarter + County	Yes	Yes	Yes
Cluster		Lender	

Notes: This table reports the 7v12 binary difference-in-differences estimate of the January 2014 collateral valuation reform on loan spreads, estimated separately by loan-size range. The dependent variable is the loan spread in percentage points, defined as the contract interest rate minus the maturity-matched Treasury yield. Column (1) restricts to loans between \$350K and \$1M ($N = 24{,}851$). Column (2) restricts to loans above \$1M through \$2M ($N = 11{,}068$). Column (3) restricts to loans above \$2M ($N = 6{,}888$). Treated states are seven states with unlimited homestead exemptions; control states are twelve low-exemption states (exemption ≤ \$23K). All specifications include controls for log loan amount, term, fixed-rate indicator, relationship indicator, previous SBA borrower indicator, collateral pledged indicator, corporation indicator, franchise indicator, and revolver indicator. Fixed effects: quarter, lender, NAICS two-digit industry × quarter, and county. Standard errors in parentheses are clustered at the lender level. The sample comprises SBA 7(a) Standard Guaranty loans, 2011Q1–2019Q4. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 8: Continuous Spread DiD Excluding Ohio

	(1) Excl. Ohio
log(effective exemption) × Post	-0.0160^{**} (0.0070)
Observations	99,887

Notes: Continuous difference-in-differences estimate of the January 2014 collateral valuation reform on loan spreads, estimated on the all-states sample with Ohio excluded ($N = 99{,}887$). The dependent variable is the loan spread in percentage points; the regressor of interest is log(effective exemption) × Post. Ohio is excluded because it raised its homestead exemption from roughly \$24,000 to \$125,000 effective March 2013, about nine months before the reform. The estimate is essentially unchanged from the all-states baseline of -1.47 bp (Table 6, column 1), confirming that the Ohio exemption change does not drive the continuous gradient. Controls and fixed effects match the baseline continuous specification (log amount, term, fixed-rate indicator, relationship, previous SBA, collateral, corporation, franchise, and revolver indicators; lender, NAICS2 × quarter, and county fixed effects). Standard errors in parentheses are clustered at the lender level. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 9: Falsification Tests and No-Reallocation

	Panel A	Panel B		
	(1) Repeat share state-qtr	(2) P(switch) >350K	(3) P(fixed) >350K	(4) Express spread Express, >$150K
Treated × Post	0.0033	−0.0147	−0.0136	0.0351
	(0.0113)	(0.0328)	(0.0223)	(0.1063)
Observations	683	5,470	43,770	9,706
Lender FE	No	Yes	Yes	Yes
Quarter/Entry FE	Yes	Yes	Yes	Yes
State FE	Yes	No	No	No

Notes: Panel A reports the state-quarter DiD on the share of repeat SBA borrowers. Panel B reports loan-level no-reallocation placebos: column (2) probability of switching from Standard to Express (estimation sample restricted to first-loan Standard >$350K borrowers); column (3) probability of fixed-rate origination on the >$350K Standard Guaranty sample; column (4) spread DiD on Express loans >$150K (Express loans face minimal collateral requirements, so the SOP collateral-valuation reform should not move spreads; the Oct 2013 fee waiver applied to Express ≤$150K, motivating the size restriction). $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 10: Design 2 Summary Statistics: UCC-Matched Sample

	FL (Treated)		CA+CO (Control)		Norm.
	Pre	Post	Pre	Post	Diff.
Panel A: Loan characteristics					
Loan amount ($1,000)	1,143.97	1,218.82	1,063.01	1,212.69	0.092
	(935.15)	(991.05)	(825.73)	(990.72)	
Term (months)	203.29	202.75	212.93	208.09	-0.105
	(91.32)	(90.34)	(93.01)	(92.16)	
Spread (pp)	4.97	4.91	4.96	4.58	0.005
	(1.09)	(1.06)	(1.01)	(1.12)	
Fixed rate	0.262	0.205	0.160	0.222	0.251
Relationship	0.048	0.066	0.088	0.082	-0.159
Previous SBA	0.101	0.111	0.167	0.156	-0.195
Collateral	0.927	0.929	0.918	0.922	0.036
Corporation	0.966	0.970	0.853	0.888	0.402
Franchise	0.094	0.185	0.075	0.149	0.069
Observations	1,143	4,284	2,818	10,135	
Panel B: Collateral characteristics (from UCC match)					
Blanket lien	0.840	0.824	0.847	0.775	-0.020
Scope score	2.51	2.66	1.97	1.80	0.453
	(1.31)	(1.42)	(1.08)	(1.22)	
Specificity score	0.70	0.83	1.22	1.17	-0.405
	(0.83)	(0.81)	(1.62)	(1.16)	
Blanket scope	0.696	0.735	0.240	0.409	1.028
Effort index	-0.35	-0.38	0.44	0.08	-0.820
	(0.77)	(0.81)	(1.12)	(1.00)	
N categories	5.09	4.88	4.83	4.42	0.117
	(2.39)	(2.46)	(1.94)	(2.19)	
Has equipment	0.862	0.815	0.878	0.832	-0.048
Has inventory	0.762	0.750	0.773	0.765	-0.027
Fixtures/RE-related	0.184	0.120	0.055	0.039	0.404
PMSI language	0.018	0.019	0.024	0.015	-0.040
Observations	1,143	4,284	2,818	10,135	

Notes: Panel A reports loan-level characteristics for all >$350K Standard Guaranty loans with UCC-classified collateral in FL (treated, unlimited homestead exemption) vs. CA+CO (control, limited exemption), 2011Q1–2019Q4. Panel B reports collateral characteristics from matched UCC filings. Effort index is computed on the full UCC-classified sample as the double-standardized sum of $(1-$ blanket scope), specificity score, and N categories. Normalized differences computed as $(\bar{x}_T - \bar{x}_C)/\sqrt{(s_T^2 + s_C^2)/2}$ using pre-reform observations.

Table 11: Collateral Composition DiD: FL vs CA+CO Around Jan 2014 Reform

	UCC Filing		Blanket Lien		N Categories		Scope Score		Equipment		Fixtures/RE-related	
	(1) Binary	(2) Cont.	(3) Binary	(4) Cont.	(5) Binary	(6) Cont.	(7) Binary	(8) Cont.	(9) Binary	(10) Cont.	(11) Binary	(12) Cont.
DiD	-0.1375^{***}	-0.0377^{***}	0.0542^{**}	0.0143^{**}	0.1461	0.0350	0.2763^{***}	0.0580^{***}	0.0197	0.0040	-0.0325	-0.0073
	(0.0255)	(0.0059)	(0.0248)	(0.0059)	(0.1599)	(0.0402)	(0.0888)	(0.0219)	(0.0241)	(0.0054)	(0.0257)	(0.0061)
Observations	29,618	29,618	18,341	18,341	18,341	18,341	18,341	18,341	18,341	18,341	18,341	18,341
Within R^2	0.0662	0.0709	0.0066	0.0070	0.0039	0.0039	0.0044	0.0049	0.0050	0.0050	0.0143	0.0143
Controls	Yes											
Lender FE	Yes											
Quarter FE	Yes											
Industry FE	Yes											
County FE	Yes											
Cluster	Lender											

Notes: Binary DiD uses $\text{Treated}_{FL} \times \text{Post}_{2014}$; Continuous uses $\log(\text{effective exemption}) \times$ Post. Cols (1)–(2): dependent variable is an indicator for having a UCC filing (extensive margin); sample is all $>$ \$350K Standard Guaranty loans in CA, FL, CO ($N = 29{,}618$). Cols (3)–(12): dependent variable is a collateral characteristic; sample restricted to UCC-classified loans ($N = 18{,}341$). Controls: log amount, term, fixed-rate, relationship, previous SBA, corporation, franchise (plus collateral indicator for cols 3–12). FE as shown. SE clustered at lender level. ***p<0.01; **p<0.05; *p<0.10.

Table 12: Soft-Information Effort DiD: Full UCC-Classified Sample

	Blanket Scope		Specificity		Effort Index	
	(1) Binary	(2) Cont.	(3) Binary	(4) Cont.	(5) Binary	(6) Cont.
DiD	-0.1298^{**}	-0.0349^{**}	0.1762^{***}	0.0364^{**}	0.3003^{***}	0.0736^{***}
	(0.0635)	(0.0164)	(0.0650)	(0.0160)	(0.0877)	(0.0245)
Observations	18,341	18,341	18,341	18,341	18,341	18,341
Within R^2	0.0062	0.0083	0.0017	0.0021	0.0045	0.0047
Controls	Yes					
Lender FE	Yes					
Quarter FE	Yes					
Industry FE	Yes					
County FE	Yes					
Cluster	Lender					

Notes: Sample: full UCC-classified, CA+FL+CO, $>$ \$350K Standard Guaranty. Effort Index = standardized composite of (1−blanket scope) + specificity score + N categories. Binary DiD uses $\text{Treated}_{FL} \times \text{Post}_{2014}$; Continuous uses log(effective exemption) × Post. SE clustered at lender. ***p<0.01; **p<0.05; *p<0.10.

Table 13: Information-Cost Pass-Through: Information Acquisition and Spread Changes (FL vs CA+CO)

	(1) Baseline	(2) Blanket × DiD	(3) Effort Change × DiD
Treated$_{FL}$ × Post	0.1289*	0.2391	0.1968**
	(0.0679)	(0.1560)	(0.0903)
Treated$_{FL}$ × Post × Blanket		−0.1392	
		(0.1255)	
Treated$_{FL}$ × Post × High Effort Δ			−0.2066
			(0.1313)
Observations	8,762	8,762	8,389
Sample	Multi-state lenders, \$350K–\$1M, UCC-matched		
Controls	Yes	Yes	Yes
FE: Quarter + Lender + Industry×Quarter + County	Yes	Yes	Yes
Cluster	Lender		

Notes: Dependent variable: loan spread (pp). Sample: UCC-matched Standard Guaranty loans, \$350K–\$1M, multi-state lenders, FL vs CA+CO, 2011Q1–2019Q4. Col (1): baseline FL × Post. Col (2): triple with blanket-scope indicator. Col (3): triple with high filing-intensity change (above-median pre-to-post lender filing intensity index shift). Controls: log amount, term, fixed rate, relationship, previous SBA, collateral, corporation, franchise, revolver. FE: quarter, lender, NAICS2 × quarter, county. SE clustered at lender level. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 14: Spread × Collateral Effort: Triple Interaction and Subsample Splits

	(1) Triple	(2) High Effort	(3) Low Effort
Treated × Post × High Effort	0.0168		
	(0.0858)		
Treated × Post	0.0757		
	(0.0566)		
Treated × Post		0.1179	
		(0.0845)	
Treated × Post			0.0093
			(0.0631)
Observations	14,688	7,226	7,402
Within R^2	0.4358	0.4531	0.4134
Controls		Yes	
Lender FE		Yes	
Quarter FE		Yes	
Industry FE		Yes	
Cluster		Lender	

Notes: Dependent variable: spread (pp). Sample: blanket lien filings with non-missing effort index, CA+FL+CO, $>$ \$350K Standard Guaranty. High Effort = effort index $>$ 0 (above median). Col (1): triple interaction; Cols (2)–(3): separate regressions by effort group. Controls: log amount, term, fixed-rate, relationship, previous SBA, collateral, corporation, franchise. FE and clustering as shown. ***p<0.01; **p<0.05; *p<0.10.

Table 15: Collateral Requirement Shift and Loan Counts by Homestead Exemption Regime

	(1)	(2)	(3)	(4)
	Pct. with Collateral		Log Loan Count	
	7v12 Binary	Continuous	7v12 Binary	Continuous
Treated × Post	-0.0491^{**}		0.0366	
	(0.0190)		(0.0927)	
log(effective exemption) × Post		-0.0072^{**}		0.0026
		(0.0032)		(0.0142)
State × Quarter Observations	683	1753	683	1753
State + Quarter FE	Yes	Yes	Yes	Yes
Cluster		State		

Notes: Unit of observation: state-quarter. Cols (1)–(2): fraction of >\$350K Standard Guaranty loans with collateral pledged. Cols (3)–(4): log loan count. Cols (1),(3): 7v12 binary (7 unlimited vs 12 low-exemption states). Cols (2),(4): continuous $\log(1+\text{HE})$ across all states. State and quarter FE. SE clustered at state level. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

A Model Derivations

This appendix micro-founds the selection cost of Section 3, states the assumptions, solves the lender's problem in closed form, and proves Propositions 1 and 2.

A.1 The supply of pledgeable borrowers

The convex selection cost is not assumed; it aggregates individual borrowers' decisions to pledge or to keep the homestead shield. Granting a consensual home lien waives the protection the exemption would otherwise provide against involuntary creditors, so borrower i forfeits a shield worth

$$\rho_i \;=\; v_i\, h, \tag{IA.1}$$

the product of an idiosyncratic exposure $v_i \geq 0$ to involuntary-creditor risk and the exemption h. The borrower consents if and only if the lender's inducement b covers the forfeited shield, $b \geq \rho_i$. Let exposure be uniform in the pledgeable population, $v_i \sim U[0,\, 1/\kappa]$, so reservation inducements $\rho_i = v_i h$ are distributed $U[0,\, h/\kappa]$ and the mass willing to pledge at inducement b is

$$S(b;h) \;=\; \Pr(\rho_i \leq b) \;=\; \frac{\kappa}{h}\, b, \qquad b \in [0,\, h/\kappa]. \tag{IA.2}$$

Inverting, a lender that brings a mass s of pledgers onto its portfolio pays the marginal inducement $b(s;h) = (h/\kappa)\, s$, which rises in s (deeper selection reaches borrowers who value the shield more) and in h (every reservation scales with the exemption). Total selection cost is the area under the inverse supply schedule,

$$C(s;h) \;=\; \int_0^s b(\sigma;h)\, \mathrm{d}\sigma \;=\; \tfrac{1}{2}\,\frac{h}{\kappa}\, s^2 \;\equiv\; \tfrac{1}{2}\,\phi(h)\, s^2, \qquad \phi(h) \equiv \frac{h}{\kappa}, \tag{IA.3}$$

convex in s and increasing in h, with both properties derived rather than asserted: convexity is the upward slope of supply, and the h-dependence is the scaling of every reservation by the exemption. The Texas corner is the limit $h \to \infty$, where every reservation diverges and selection vanishes.

Remark 1 (Only the magnitudes need the uniform). The uniform distribution is used only for the closed-form curvature $\phi(h) = h/\kappa$. Every sign result below requires only that reservation inducements be increasing in h: for any distribution of v_i with $\rho_i = v_i h$ rising in h, the inverse supply $b(s;h)$ increases in s and h, the cost $C(s;h)$ is convex in s and increasing in h, and hence $\partial s^*/\partial h < 0$, $\mathrm{sign}(\partial a^*/\partial h) = -\,\mathrm{sign}(\gamma_0)$, and the diversification-versus-concentration boundary are unchanged.

A.2 Assumptions

Assumption 1 (Binding asset-class over-crediting)**.** *$\mu_m > 0$ on movable collateral while $\mu_{RE} \approx 0$ on real estate, and the rule binds because rejected borrowers re-shop. The asymmetry $\mu_m > \mu_{RE}$, not over-crediting per se, gives selection (escape) and enforcement (offset) distinct roles.*

Assumption 2 (Uniform pari-passu retention)**.** *The guarantor shares every recovery channel proportionally, so the lender retains the same fraction $\lambda \in (0, 1]$ of real-estate and of movable recovery. λ enters only through $m = \lambda\mu_m$ and does not differentially attenuate one margin.*

Assumption 3 (Pledge-supply relevance)**.** *A non-trivial mass of borrowers will consent to a home lien at the prevailing terms, so the exemption is interior ($h < \infty$) and $\phi(h) = h/\kappa$ is finite. Where this fails ($h \to \infty$ or the Texas bar), selection is prohibitively costly and only the enforcement margin operates.*

Assumption 4 (Enforcement targets movable assets)**.** *Enforcement perfects liens on movable business collateral, the class the rule over-credits and the homestead neither shields nor governs the supply of. Enforcement cost and recovery are independent of h except through the portfolio-composition channel $\vartheta_0 + \gamma_0 s$.*

Assumption 5 (Bounded complementarity)**.** *$m^2\gamma_0^2 < \phi(h_{\min})\,\psi$ at the lowest exemption $h_{\min}$ in the support. Since $\phi(h)$ rises in h, this second-order condition then holds throughout the support, guaranteeing a unique interior optimum.*

A.3 The joint optimum

Lemma 1 (Optimal selection and enforcement)**.** *Under Assumption 5 the payoff (1) is strictly concave with a unique interior optimum solving*

$$\phi(h)\, s^* = m\,(1 + \gamma_0 a^*), \qquad \psi\, a^* = m\,(\vartheta_0 + \gamma_0 s^*). \tag{IA.4}$$

Writing $D(h) \equiv \phi(h)\psi - m^2\gamma_0^2 > 0$ and $N \equiv m(\psi + m\gamma_0\vartheta_0)$,

$$s^* = \frac{N}{D(h)}, \qquad a^* = \frac{m\,\phi(h)\,\vartheta_0 + m^2\gamma_0}{D(h)}. \tag{IA.5}$$

We maintain an interior optimum with $s^* > 0$ and $a^* \geq 0$, equivalently $N = m(\psi + m\gamma_0\vartheta_0) > 0$ and $m\phi(h)\vartheta_0 + m^2\gamma_0 \geq 0$; these are Assumption 3 expressed in coordinates. In the diversification *regime ($\gamma_0 < 0$) the second condition is $\phi(h) \geq m|\gamma_0|/\vartheta_0$, so $a^* \geq 0$ holds for exemptions that are not too low; the comparative-static signs below are stated for this interior region. The gradient signs are global properties of the interior solution: the substitution reading concerns the sign of $\partial a^*/\partial h$, not a guarantee of strictly positive enforcement at every exemption.*

Proof. The Hessian of (1) is $\left(\begin{smallmatrix} -\phi(h) & m\gamma_0 \\ m\gamma_0 & -\psi \end{smallmatrix}\right)$, negative definite iff $\phi(h), \psi > 0$ and $\phi(h)\psi - m^2\gamma_0^2 > 0$. The latter is slackest at $h_{\min}$, which is Assumption 5; for $h > h_{\min}$ the curvature $\phi(h)$ only grows, so concavity holds throughout. The first-order conditions $\partial_s \Pi = m(1 + \gamma_0 a) - \phi(h)s = 0$ and $\partial_a \Pi = m(\vartheta_0 + \gamma_0 s) - \psi a = 0$ are (IA.4). Solving the linear system $\left(\begin{smallmatrix} \phi(h) & -m\gamma_0 \\ -m\gamma_0 & \psi \end{smallmatrix}\right)\left(\begin{smallmatrix} s \\ a \end{smallmatrix}\right) = \left(\begin{smallmatrix} m \\ m\vartheta_0 \end{smallmatrix}\right)$ by Cramer's rule gives (IA.5). $\square$

Because λ enters only through $m = \lambda\mu_m$, it scales both s^* and a^* but divides out of every ratio and sign below: pari-passu sharing rescales the level of defensive activity without changing its composition.

A.4 Homestead gradients

The exemption enters only through $\phi(h) = h/\kappa$, with $\partial\phi/\partial h = 1/\kappa > 0$.

Proof of Proposition 1. Write $s^* = N/D(\phi)$ with $D(\phi) = \phi\psi - m^2\gamma_0^2$ and N constant in ϕ. Then $\partial s^*/\partial\phi = -N\,D'(\phi)/D^2 = -N\psi/D^2 < 0$, since $N, \psi > 0$ at an interior optimum. Multiplying by $\partial\phi/\partial h = 1/\kappa > 0$ gives $\partial s^*/\partial h = -N\psi/(\kappa D^2) < 0$, independent of γ_0. $\square$

Proof of Proposition 2. Write $a^* = A(\phi)/D(\phi)$ with $A(\phi) = m\vartheta_0\phi + m^2\gamma_0$, so $A' = m\vartheta_0$ and $D' = \psi$. Then $\partial a^*/\partial\phi = (A'D - AD')/D^2$, whose numerator is $m\vartheta_0(\phi\psi - m^2\gamma_0^2) - (m\vartheta_0\phi + m^2\gamma_0)\psi = -m^2\gamma_0(\psi + m\gamma_0\vartheta_0)$, the $m\vartheta_0\phi\psi$ terms cancelling. Since $\psi + m\gamma_0\vartheta_0 = N/m > 0$, the sign is $-\operatorname{sign}(\gamma_0)$; multiplying by $1/\kappa > 0$ preserves it. $\square$

The ratio of gradients is

$$\frac{\partial a^*/\partial h}{\partial s^*/\partial h} = \frac{m^2\gamma_0(\psi + m\gamma_0\vartheta_0)}{N\psi} = \frac{m\gamma_0}{\psi}, \tag{IA.6}$$

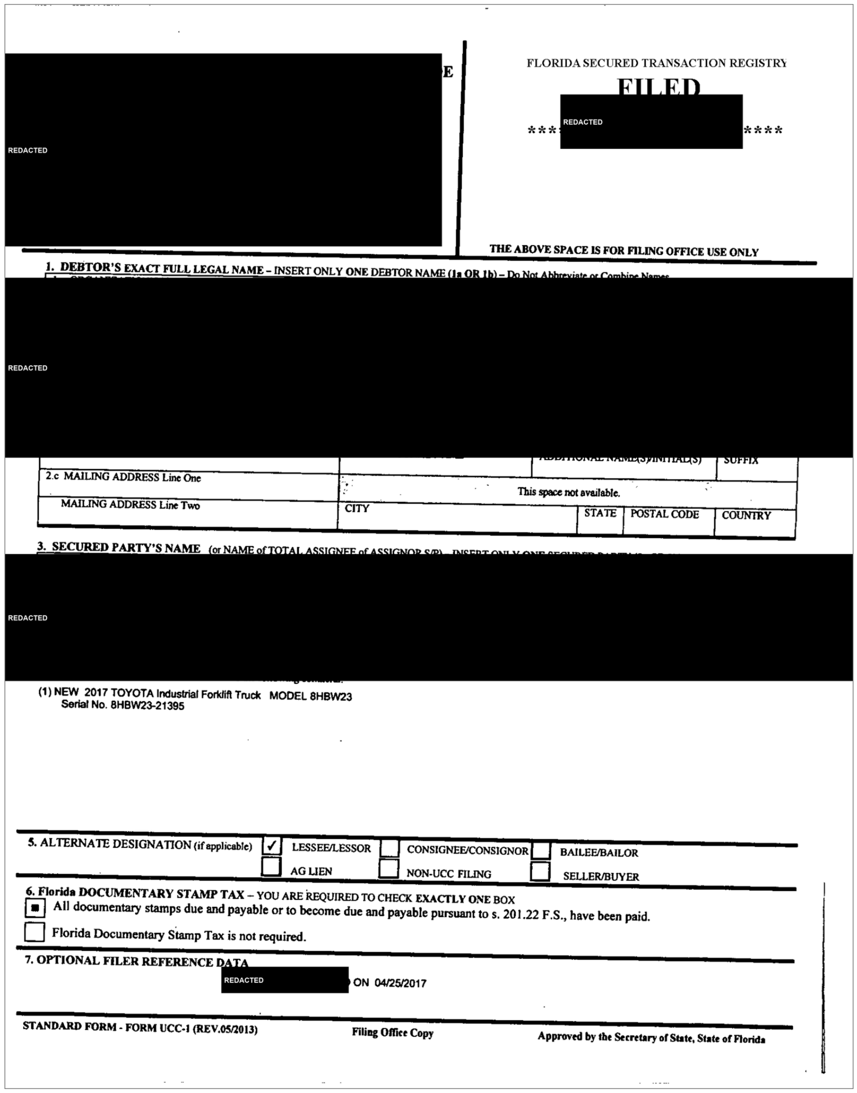
so the diversification-versus-concentration boundary sits at $\operatorname{sign}(\gamma_0)$ alone, with the $(\psi + m\gamma_0\vartheta_0)$ and $1/\kappa$ factors cancelling. Corollary 1 follows.

A.5 Scope

The model is partial equilibrium: the representative lender takes the supply schedule (IA.2) and the marginal inducement $b(s; h) = \phi(h)s$ as given. The participation pool is a common resource, so a full equilibrium would make ϕ an equilibrium scarcity, adding a congestion externality and a fixed point in the price of pledging. Because congestion acts through the same curvature ϕ that carries the exemption, it would rescale the level of the inducement faced by all lenders but leave the sign of every homestead gradient and the diversification-versus-concentration boundary, which depend only on $\mathrm{sign}(\gamma_0)$, untouched. The present version isolates the within-lender optimization, which is sufficient for the two gradients and the dichotomy.

B An Example UCC Financing Statement

The collateral-description measures used in the collateral-specificity analysis are built from the free-text collateral field of the UCC-1 financing statement, the public filing a lender makes to perfect its security interest in a borrower's movable assets. The example below reproduces one such filing from the matched Florida sample.



[redacted] REDACTED

FLORIDA SECURED TRANSACTION REGISTRY

FILED

*** [redacted] REDACTED ****

THE ABOVE SPACE IS FOR FILING OFFICE USE ONLY

1. DEBTOR'S EXACT FULL LEGAL NAME - INSERT ONLY ONE DEBTOR NAME (1a OR 1b) - Do Not Abbreviate or Combine Names

[redacted] REDACTED

SUFFIX

2.c MAILING ADDRESS Line One

This space not available.

MAILING ADDRESS Line Two | CITY | STATE | POSTAL CODE | COUNTRY

3. SECURED PARTY'S NAME (or NAME of TOTAL ASSIGNEE of ASSIGNOR S/P)

[redacted] REDACTED

(1) NEW 2017 TOYOTA Industrial Forklift Truck MODEL 8HBW23
Serial No. 8HBW23-21395

5. ALTERNATE DESIGNATION (if applicable) ✓ LESSEE/LESSOR ☐ CONSIGNEE/CONSIGNOR ☐ BAILEE/BAILOR ☐ AG LIEN ☐ NON-UCC FILING ☐ SELLER/BUYER

6. Florida DOCUMENTARY STAMP TAX – YOU ARE REQUIRED TO CHECK EXACTLY ONE BOX

■ All documentary stamps due and payable or to become due and payable pursuant to s. 201.22 F.S., have been paid.

☐ Florida Documentary Stamp Tax is not required.

7. OPTIONAL FILER REFERENCE DATA

[redacted] REDACTED ON 04/25/2017

STANDARD FORM - FORM UCC-1 (REV.05/2013) Filing Office Copy Approved by the Secretary of State, State of Florida

Figure IA.1: An example UCC-1 financing statement from the matched Florida sample. The form records, in numbered boxes, the debtor (the borrowing firm, box 1), the secured party (the lender, box 3), and a free-text collateral description (box 4); box 5 designates the filing type and box 6 the Florida documentary-stamp status. The debtor, secured-party, and contact identifiers are redacted; the collateral description is reproduced as filed. In this filing the lender records a single titled asset by make, model, and serial number, an itemized, asset-specific description. The financing statement perfects the lien and is the source of the blanket-scope, specificity, and filing-intensity measures used in the collateral-specificity results.

Across filings, the recorded collateral language ranges from generic blanket liens, which claim all assets in a single phrase, to itemized descriptions that name individual assets by make, model, and serial or VIN number. The following examples illustrate this range and the inputs it provides to the filing-intensity measures.

Table IA.1: Collateral descriptions across the specificity spectrum

Collateral description (anonymized)	Blanket lien	Asset identifiers	Primary asset class
"All assets."	Yes	0	All categories
"One (1) [make] truck, model #[$\cdots$], serial #[$\cdots$]."	No	2	Vehicles
"(1) 2011 [make] rough-terrain crane, VIN [$\cdots$]" (three identical units)	No	4	Equipment

Notes: Three example filings from the matched California, Florida, and Colorado sample, reproduced verbatim and anonymized by masking the manufacturer model and the serial or VIN digits. "Blanket lien" is the blanket-scope indicator (one for filings that claim all assets, zero otherwise). "Asset identifiers" counts the asset-specific identifiers (serial or VIN numbers) in the collateral description. These inputs feed the specificity score and the composite filing-intensity index defined in Section 4.2.

C Additional Internet Appendix Figures and Tables

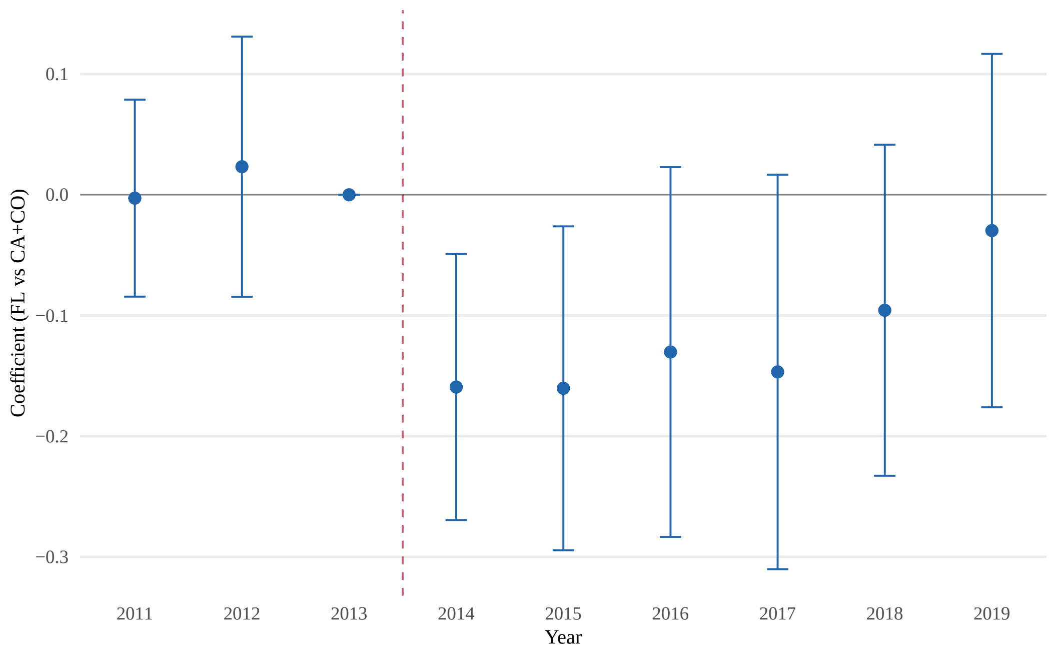


Figure IA.2: Event Study: Blanket-Scope Indicator, Full UCC-Classified Sample. Coefficients on Year $\times$ Treated$_{FL}$ interactions from equation (4) with the blanket-scope indicator as outcome, estimated on the full UCC-classified sample. Reference year is 2013. Bars show 95% confidence intervals clustered at the lender level. Wald test for pre-trends: $F = 0.16$, $p = 0.854$.

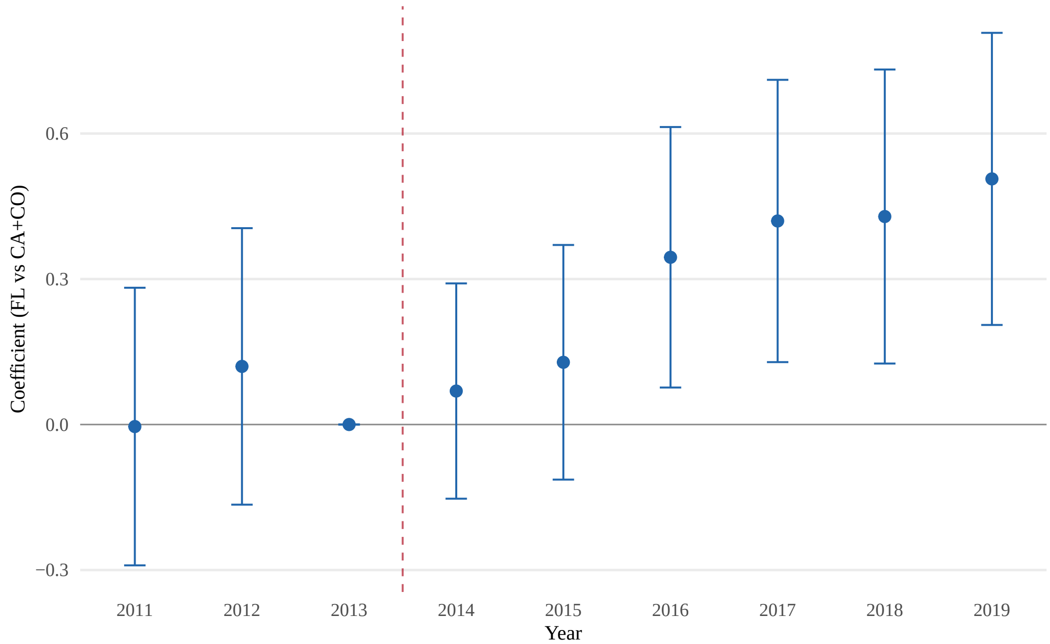


Figure IA.3: Event Study: Collateral Scope Score (All UCC-Classified). Coefficients on Year × Treated$_{FL}$ interactions with scope score as outcome, estimated on the full UCC-classified sample. Reference year is 2013. Wald test for pre-trends: $F = 0.69$, $p = 0.504$.

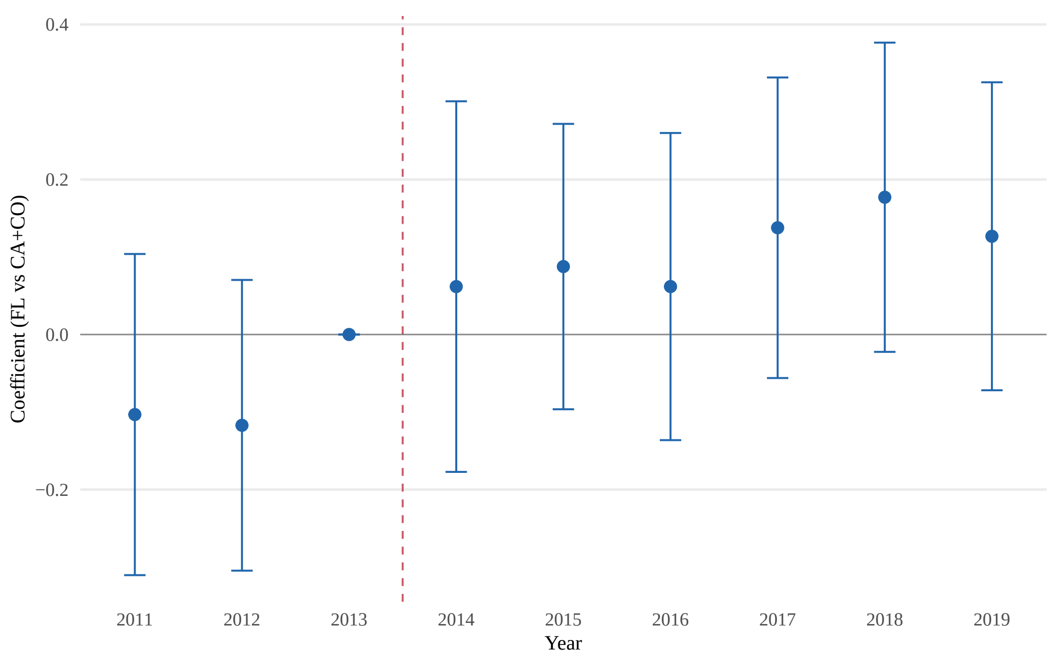


Figure IA.4: Event Study: Specificity Score, Full UCC-Classified Sample. Coefficients on Year × Treated$_{FL}$ interactions with specificity score as outcome, estimated on the full UCC-classified sample. Reference year is 2013. Wald test for pre-trends: $F = 0.78$, $p = 0.461$.

Table IA.2: Specificity Gradient Under Match-Selection Corrections

	(1) Baseline	(2) IP match-prob. weighted	(3) Within-collateral-type FE
Treated$_{FL}$ $\times$ Post	0.1762^{***}	0.1771^{***}	0.1626^{**}
	(0.0650)	(0.0652)	(0.0700)
Observations	18,341	18,341	18,341
Controls	Yes	Yes	Yes
Lender / quarter / NAICS / county FE	Yes	Yes	Yes
Inverse match-prob. weights	No	Yes	No
Collateral-type FE	No	No	Yes

Notes: Robustness of the filing-specificity DiD (the specificity score on Treated$_{FL}$ $\times$ Post) to selection into the UCC-matched sample, whose match rate is itself treatment-affected. Column (1) is the baseline estimate of equation (4) on the full UCC-classified sample. Column (2) weights by the inverse estimated match probability. Column (3) adds within-collateral-type fixed effects. The gradient is essentially unchanged under both corrections, which address selection on observable characteristics into the matched sample. Standard errors in parentheses are clustered at the lender level. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table IA.3: Lender-Continuity Robustness: Continuous UCC Filers

	Full sample (1)	Continuous filers (2)
Blanket-scope indicator	-0.1298^{**}	-0.1273^{**}
	(0.0635)	(0.0633)
Specificity score	0.1762^{***}	0.1639^{**}
	(0.0650)	(0.0653)
Filing-intensity index	0.3003^{***}	0.2878^{***}
	(0.0877)	(0.0852)
Observations	18,341	17,716
Lenders	185	141
Controls	Yes	Yes
Lender / quarter / NAICS / county FE	Yes	Yes

Notes: The three intensive-margin collateral-adaptation outcomes (blanket-scope indicator, specificity score, filing-intensity index) estimated on the full UCC-classified sample (column 1) and restricted to the 141 lenders that filed at least one UCC statement in both the pre- and post-reform periods (column 2, N = 17,716), addressing the concern that the headline pattern reflects compositional entry of new UCC filers rather than incumbent behavioral change. All three coefficients move by no more than about seven percent and retain significance. Specification matches equation (4); standard errors in parentheses are clustered at the lender level. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table IA.4: Summary Statistics: Binary HE Design (7v12 States)

	Treated (Unlimited HE) Pre	Treated (Unlimited HE) Post	Control (Low HE) Pre	Control (Low HE) Post	Norm. Diff.
Panel A: All loans (>$350K Standard Guaranty)					
Loan amount (in $1,000)	1171.74	1290.29	1142.41	1246.63	0.044
	(925.08)	(1045.34)	(892.09)	(1012.33)	
Term (months)	206.39	207.78	204.78	204.74	0.030
	(90.19)	(90.58)	(90.95)	(88.94)	
Spread (pp)	5.02	4.71	5.18	4.89	-0.178
	(1.05)	(1.10)	(0.95)	(1.04)	
Fixed rate	0.230	0.223	0.157	0.178	0.132
Relationship	0.072	0.066	0.075	0.070	-0.017
Previous SBA	0.120	0.107	0.128	0.123	-0.043
Collateral	0.931	0.908	0.878	0.917	0.026
Corporation	0.905	0.934	0.929	0.944	-0.051
Franchise	0.107	0.218	0.116	0.209	0.021
Observations	4,651	15,066	6,588	17,729	
Panel B: Repeat borrowers ($350K–$1M)					
First loan amount (in $1,000)	606.71	608.84	598.76	616.34	-0.006
	(183.00)	(184.93)	(177.32)	(180.70)	
First loan term (months)	171.19	180.83	185.06	182.20	-0.068
	(91.08)	(91.64)	(91.13)	(87.56)	
First loan spread (pp)	5.02	4.93	5.32	5.06	-0.198
	(1.05)	(1.09)	(0.83)	(1.01)	
Fixed rate	0.227	0.191	0.108	0.177	0.146
Relationship	0.136	0.218	0.209	0.192	-0.028
Previous SBA	0.199	0.266	0.313	0.261	-0.094
Collateral	0.920	0.915	0.824	0.897	0.161
Corporation	0.926	0.935	0.942	0.963	-0.099
Franchise	0.045	0.212	0.097	0.182	0.004
Number of loans	2.17	2.09	2.11	2.09	0.049
	(0.55)	(0.35)	(0.39)	(0.32)	
Observations	176	293	278	406	

Notes: Panel A reports loan-level statistics for all >$350K Standard Guaranty loans in 7 unlimited-homestead-exemption states (treated) vs. 12 low-exemption states (control), 2011Q1–2019Q4. Panel B reports first-loan characteristics for repeat borrowers ($350K–$1M). Normalized differences computed as $(\bar{x}_T - \bar{x}_C)/\sqrt{(s_T^2 + s_C^2)/2}$.

Table IA.5: Spread Difference-in-Differences: Binary HE Design (7v12 States)

	Panel A: Full Sample			Panel B: By Loan Size		
	(1) Baseline	(2) Additive FE	(3) + HPrice	(4) \$350K–500K	(5) \$500K–1M	(6) \$1M–2M
Treated × Post	−0.1072***	−0.1063***	−0.1095***	−0.1425**	−0.1021**	−0.0898*
	(0.0389)	(0.0405)	(0.0386)	(0.0563)	(0.0421)	(0.0467)
Observations	43,733	43,770	43,171	9,095	15,367	11,068
Within R^2	0.3816	0.3828	0.3802	0.3971	0.3873	0.3423
Lender FE	Yes	Yes	Yes	Yes	Yes	Yes
NAICS2 × Quarter FE	Yes	No	Yes	Yes	Yes	Yes
Quarter FE (additive)	No	Yes	No	No	No	No
County FE	Yes	Yes	Yes	Yes	Yes	Yes
Home price control	No	No	Yes	No	No	No

Notes: Dependent variable is the loan spread (percentage points over matched-maturity Treasury). Treated = 7 unlimited homestead exemption states; Control = 12 low-exemption states. Sample: >\$350K Standard Guaranty loans, 2011Q1–2019Q4. Column (1) is the preferred specification with NAICS2 × quarter fixed effects. Column (2) substitutes additive quarter FE as a robustness check. Columns (3)–(6) all use the preferred NAICS2 × quarter specification: column (3) adds a county-level home price control; columns (4)–(6) restrict by loan-amount tier. All columns include controls for log loan amount, term, fixed-rate indicator, relationship lending, previous SBA, collateral, corporation, revolver, and franchise indicators. Standard errors clustered at the lender level. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table IA.6: Pre-Trend Tests

	Panel A: Repeat Ratio		Panel B: Spread	
	Coef.	SE	Coef.	SE
2011	0.0018	(0.0136)	0.0129	(0.0320)
2012	0.0415**	(0.0193)	0.0577*	(0.0310)
2014	0.0169	(0.0222)	−0.0476	(0.0376)
2015	0.0179	(0.0212)	−0.0684	(0.0422)
2016	0.0270*	(0.0148)	−0.0818*	(0.0493)
2017	0.0258	(0.0159)	−0.0664	(0.0500)
2018	0.0283	(0.0174)	−0.1149**	(0.0554)
2019	−0.0086	(0.0217)	−0.1331**	(0.0654)
Pre-trend Wald F	2.54		1.96	
p-value	0.080		0.141	

Notes: Event study coefficients with 2013 as base year. Panel A: any-repeat share at state-quarter level. Panel B: loan-level spread. Pre-trend Wald test: joint significance of 2011 and 2012 coefficients. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table IA.7: Leave-One-State-Out Robustness

Dropped State	Coef.	SE	N
AL	-0.1037^{**}	(0.0415)	42,841
AR	-0.1005^{**}	(0.0407)	43,132
FL	-0.1653^{***}	(0.0507)	37,433
GA	-0.1073^{**}	(0.0435)	38,218
IA	-0.1037^{**}	(0.0413)	43,367
IL	-0.1034^{***}	(0.0386)	40,211
IN	-0.1120^{***}	(0.0404)	42,033
KS	-0.1078^{**}	(0.0419)	43,101
KY	-0.1031^{**}	(0.0404)	43,067
MD	-0.1045^{**}	(0.0406)	42,514
MO	-0.1100^{***}	(0.0418)	42,193
NJ	-0.1127^{***}	(0.0418)	40,621
OK	-0.1048^{**}	(0.0410)	42,958
PA	-0.0920^{**}	(0.0399)	41,024
SD	-0.1056^{***}	(0.0405)	43,623
TN	-0.1040^{**}	(0.0417)	42,507
TX	-0.0141	(0.0430)	33,154
VA	-0.1024^{**}	(0.0428)	42,219
WY	-0.1052^{**}	(0.0409)	43,618

Notes: Each row drops one state from the 7v12 binary sample and re-estimates the loan-level spread DiD. Sample: >\$350K Standard Guaranty loans. FE: quarter, lender, NAICS2, county. SE clustered at lender level. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table IA.8: Continuous Leave-One-State-Out: Spread

State	Coef.	SE	p	N	State	Coef.	SE	p	N
AL	-0.0145^{**}	(0.0070)	0.039	102,578	NC	-0.0148^{**}	(0.0068)	0.030	100,656
AR	-0.0141^{**}	(0.0069)	0.041	102,869	ND	-0.0148^{**}	(0.0069)	0.032	103,390
AZ	-0.0148^{**}	(0.0069)	0.031	100,373	NE	-0.0148^{**}	(0.0069)	0.033	103,133
CA	-0.0140^{**}	(0.0070)	0.046	83,804	NH	-0.0148^{**}	(0.0069)	0.032	103,276
CO	-0.0151^{**}	(0.0069)	0.030	99,890	NJ	-0.0150^{**}	(0.0070)	0.033	100,355
CT	-0.0148^{**}	(0.0069)	0.032	102,670	NM	-0.0149^{**}	(0.0069)	0.032	102,942
DC	-0.0151^{**}	(0.0069)	0.028	103,306	NV	-0.0145^{**}	(0.0069)	0.038	102,510
DE	-0.0148^{**}	(0.0069)	0.032	103,265	NY	-0.0147^{**}	(0.0069)	0.034	100,221
FL	-0.0243^{***}	(0.0086)	0.005	97,166	OH	-0.0161^{**}	(0.0071)	0.024	99,918
GA	-0.0148^{**}	(0.0072)	0.041	97,951	OK	-0.0149^{**}	(0.0069)	0.031	102,695
IA	-0.0147^{**}	(0.0070)	0.037	103,101	OR	-0.0151^{**}	(0.0069)	0.030	101,703
ID	-0.0148^{**}	(0.0069)	0.032	102,803	PA	-0.0134^{*}	(0.0068)	0.050	100,760
IL	-0.0140^{**}	(0.0066)	0.034	99,947	RI	-0.0149^{**}	(0.0069)	0.031	103,282
IN	-0.0154^{**}	(0.0069)	0.025	101,772	SC	-0.0148^{**}	(0.0069)	0.032	102,250
KS	-0.0155^{**}	(0.0070)	0.028	102,837	SD	-0.0150^{**}	(0.0068)	0.028	103,360
KY	-0.0146^{**}	(0.0068)	0.034	102,803	TN	-0.0145^{**}	(0.0072)	0.044	102,243
LA	-0.0146^{**}	(0.0069)	0.036	102,701	TX	-0.0034	(0.0064)	0.596	92,887
MA	-0.0151^{**}	(0.0069)	0.029	102,478	UT	-0.0159^{**}	(0.0068)	0.021	101,650
MD	-0.0146^{**}	(0.0069)	0.034	102,251	VA	-0.0142^{*}	(0.0074)	0.055	101,955
ME	-0.0149^{**}	(0.0069)	0.031	103,330	VT	-0.0148^{**}	(0.0069)	0.032	103,418
MI	-0.0147^{**}	(0.0067)	0.029	100,463	WA	-0.0150^{**}	(0.0069)	0.029	99,921
MN	-0.0147^{**}	(0.0069)	0.034	101,567	WI	-0.0143^{**}	(0.0068)	0.036	101,110
MO	-0.0150^{**}	(0.0071)	0.035	101,929	WV	-0.0148^{**}	(0.0069)	0.032	103,369
MS	-0.0151^{**}	(0.0069)	0.029	102,810	WY	-0.0148^{**}	(0.0069)	0.033	103,351
MT	-0.0148^{**}	(0.0069)	0.031	103,235					

Notes: Each row drops one state from the all-states continuous specification and re-estimates the spread DiD using log(effective exemption$_s$) $\times$ Post$_t$. Sample: >$350K Standard Guaranty loans. FE: quarter, lender, NAICS2, county. SE clustered at lender level. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table IA.9: UCC Matched-Sample Composition: Pre vs Post by Regime

	FL (treated)			CA+CO (control)		
	Pre	Post	ND	Pre	Post	ND
Spread (pp)	4.97	4.91	-0.06	4.96	4.58	-0.36
Amount ($1,000)	1143.97	1218.82	0.08	1063.01	1212.69	0.16
Term (months)	203.29	202.75	-0.01	212.93	208.09	-0.05
Fixed rate (%)	26.2	20.5	—	16.0	22.2	—
Previous SBA (%)	10.1	11.1	—	16.7	15.6	—
Relationship (%)	4.8	6.6	—	8.8	8.2	—
Loans matched	1,143	4,284	—	2,818	10,135	—

Notes: Matched-sample loan-level observables for >$350K Standard Guaranty loans, by state (FL vs CA+CO) and reform period. ND = $(\bar{x}_{\text{post}} - \bar{x}_{\text{pre}})/\sqrt{(s^2_{\text{pre}} + s^2_{\text{post}})/2}$ within each regime; columns 4 and 7 show within-regime composition shift. Small ND across both regimes indicates the matched-sample composition is stable pre-to-post within each state group, so matched-sample inference is not contaminated by selection on observables.

Table IA.10: Charge-Off Event Study

Year	Coefficient	SE
2011	-0.0185^{*}	(0.0101)
2012	-0.0003	(0.0104)
2014	-0.0022	(0.0087)
2015	0.0090	(0.0079)
2016	0.0097	(0.0085)
2017	0.0080	(0.0077)
2018	0.0097	(0.0091)
2019	0.0013	(0.0092)
Observations	43,770	

Notes: Binary charge-off event study with 2013 as base year. Sample: >$350K Standard Guaranty, 7v12 states. Lender FE + quarter FE, lender-clustered SEs. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table IA.11: Symmetric Window Robustness (2011Q1–2016Q4)

	Spread DiD			
	(1) Full	(2) \$350K–500K	(3) \$500K–1M	(4) \$1M–2M
Treated × Post	-0.1077^{***}	-0.1626^{***}	-0.1223^{***}	-0.1040^{**}
	(0.0359)	(0.0549)	(0.0446)	(0.0416)
Observations	27,599	5,826	9,717	7,076
Lender FE	Yes	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes	Yes

Notes: Symmetric 3+3 year window around the Jan 2014 reform (2011Q1–2016Q4). Loan-level spread DiD across the full sample (col 1) and three loan-size tiers (cols 2–4). All specs use lender + quarter + NAICS2 + county FE, lender-clustered SEs. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table IA.12: Wild Cluster Bootstrap Inference

Specification	$\hat{\beta}$	Lender p	Boot p	Clusters
Panel A: Design 1 — Binary 7v12				
Spread DiD baseline (T2 col 1)	-0.106	0.009	0.000	1140
Spread DiD + hprice (T2 col 2)	-0.109	0.007	0.000	1128
Panel B: Design 1 — Continuous HE				
Spread $\sim$ log(effective exemption) (T6)	-0.016	0.021	0.000	2129
Spread $\sim$ rank(ex) (T6)	-0.095	0.013	0.000	2129

Notes: Each row reports the point estimate, lender-clustered p-value, and wild cluster bootstrap p-value at the state level (Webb six-point distribution, 99,999 replications). Implemented via `fwildclusterboot`.

Table IA.13: Clustering Robustness

Specification	$\hat{\beta}$	Standard errors / p-values				Clusters		
		Lender	County	Two-way	State	G_L	G_C	G_S
Spread baseline (T2)	-0.1063^{***}	0.009	0.000	0.009	0.111	315	1387	19
		(0.0405)	(0.0222)	(0.0407)	(0.0634)			
Spread + hprice (T2)	-0.1092^{***}	0.007	0.000	0.007	0.091	315	1370	19
		(0.0402)	(0.0217)	(0.0399)	(0.0612)			

Notes: Each row re-estimates the headline specification referenced in parentheses under four clustering levels. "Lender" clusters standard errors at the lender level (baseline throughout the paper). "County" clusters at the borrower county level. "Two-way" uses multi-way clustering on lender and county simultaneously (Cameron, Gelbach, and Miller 2011). "State" clusters at the borrower state level (19 states in the 7v12 design). Standard errors in parentheses; p-values reported above. G_L, G_C, G_S denote the number of lender, county, and state clusters. Stars on $\hat{\beta}$ reflect lender-clustered significance: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table IA.14: Within-Borrower Spread DiD: Borrower Fixed Effects

	(1) Baseline	(2) +Borr FE (full)	(3) +Borr FE (qtr only)	(4) +Borr FE (+lender)	(5) +Borr FE (no ctrl)
Panel A: 7v12 Binary					
Treated × Post	-0.1072^{***}	0.0254	0.0687	0.0653	0.0211
	(0.0389)	(0.1273)	(0.0964)	(0.1092)	(0.1433)
Observations	43,733	5,557	5,800	5,761	5,800
Panel B: All-States Continuous					
Log(1+HE) × Post	-0.0147^{**}	0.0042	0.0105	0.0132	0.0060
	(0.0068)	(0.0171)	(0.0154)	(0.0169)	(0.0236)
Observations	103,476	15,102	15,267	15,235	15,267
Quarter FE	Yes	Yes	Yes	Yes	Yes
Lender FE	Yes	Yes		Yes	
NAICS2 × Qtr	Yes	Yes			
County FE	Yes	Yes			
Borrower FE		Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	

Notes: Column (1) reproduces the baseline spread DiD. Columns (2)–(5) add borrower fixed effects under varying FE saturation. The sample collapses because borrower FE absorbs the state-level treatment indicator for single-loan borrowers; only repeat borrowers with pre- and post-reform loans contribute. Standard errors clustered at the lender level. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.